DEPLOYMENT LINK

https://stock-level-prediction-and-control-2rqr.onrender.com/

R & R Apparel Sourcing

+91 421 4244889
+91 96 8889 8889

#22/12 Jeeva Colony North Extn,
Gandhinagar Post Tirupur 641 603
jayaraman@randrapparelsourcing.com

GSTIN : 33AGTPJ6636Q1ZJ

Date:
Date: 05.07.2026

TO WHOMSOEVER IT MAY CONCERN

This is to certify that Ms. Pratheeba S, Register Number 2428B0333, student of III Year B.Sc. Computer Science with Data Analytics (CSDA), KPR College of Arts, Science and Research, Coimbatore, is permitted to carry out her Capstone Project titled "Stock Level Prediction and Control Software" at R&R Apparel Sourcing, during the period from June 2026 to October 2026.

The project focuses on developing a Stock Level Prediction and Control Software to monitor inventory levels, predict future stock requirements, and support efficient stock management. The system is designed to help reduce stock shortages and overstock situations, improve inventory control, and assist in timely decision-making. All data and information collected will be used strictly for academic purposes only.

We wish her all success in completing the project.

For R&R APPAREL SOURCING

Authorised Signatory

R & R Apparel Sourcing

+91 421 4244889
+91 96 8889 8889

#22/12 Jeeva Colony North Extn.,
Gandhinagar Post Tirupur 641 603
jayaramme@randrapparelsourcing.com

GSTIN : 33AGTPJ6636Q1ZJ

CONSULTANCY OFFER LETTER

PLACE: Coimbatore **DATE:** 15.06.2026

To
The Principal
KPR College of Arts, Science and Research
Coimbatore – 641407

Through
The Head of the Department
Department of Computer Science with Data Analytics
KPR College of Arts, Science and Research
Coimbatore – 641407

Dear Sir/Madam,

Subject: Consultancy Offer for Project Development – "Stock Level Prediction and Control Software"

We are pleased to inform you that our company proposes to engage **Ms. Pratheeba S (Register No.: 2428B0333, Roll No.: 24CSD033)**, a student of III B.Sc. Computer Science with Data Analytics at KPR College of Arts, Science and Research, for consultancy services related to the development of the project titled **"Stock Level Prediction and Control Software."**

The project aims to develop a software solution for monitoring inventory levels, predicting future stock requirements, and improving inventory control through efficient stock management techniques. The system is intended to reduce stock shortages and overstock situations while supporting timely decision-making for better operational efficiency.

In this regard, we are pleased to offer a consultancy remuneration of **Rs. 2,500/- (Rupees Two Thousand Five Hundred only)** to **Ms. Pratheeba S** as a professional consultancy fee for her valuable contribution towards the successful design and development of the above-mentioned project for our company.

The consultancy engagement will be carried out under the academic guidance and approval of **Dr. M. Manoj,** Assistant Professor, Department of Computer Science with Data Analytics, KPR College of Arts, Science and Research, and is intended to provide practical exposure and industry-oriented learning while ensuring the successful completion of the project objectives.

We kindly request the Department to accord its approval and extend the necessary support to facilitate this consultancy engagement. We appreciate the continued cooperation of KPR College of Arts, Science and Research in strengthening industry–academia collaboration.

Thanking you.

Yours sincerely,
For R & R Apparel Sourcing

For R&R APPAREL SOURCING

Authorised Signatory

R & R Apparel Sourcing

+91 421 4244889
+91 96 8889 8889

#22/2 Jeeva Colony North Extn,
Gandhinagar Post Tiruppur 641 603,
jayavarman@randrapparelsourcing.com

GSTIN : 33AGTPJ6636Q1ZJ

PROJECT COMPLETION LETTER

Place: Coimbatore
Date: 06.10.2026

To
The Principal
KPR College of ArtsScience and Research
Coimbatore – 641407

Through
The Head of the Department
Department of ComputerScience withData Analytics
KPR College of Arts Science and Research
Coimbatore – 641407

Dear Sir/Madam,

Subject: Consultancy Completion Letter – Project "Stock Level Prediction and Control Software"

This is to formally inform you that **Ms. Pratheeba S,** (Reg.No: 2428B0333; Roll.No: 24CSD033), pursuing III B.Sc. Computer Science with Data Analytics at KPR College of Arts Science and Research, has successfully completed the consultancy services assigned to her for our Company in connection with the project titled **"Stock Level Prediction and Control Software."**

The consultancy work was carried out under the academic guidance and approval of **Dr.M. Manoj,** Assistant Professor, Department of Computer Science with Data Analytics, KPRCAS, and involved the design and development activities as per the project objectives mutually agreed upon. We are satisfied with the quality of work delivered and the professional approach demonstrated by the student during the consultancy period.

In recognition of her contribution, a consultancy remuneration of **Rs. 2,500/- (Rupees Two Thousand Five Hundred only)** has been paid to **Ms. Pratheeba S** towards the successful completion of the consultancy assignment.

We place on record our appreciation to the Department of Computer Science with Data Analytics, KPRCAS, for its support and cooperation in facilitating this industry–academia consultancy engagement. We look forward to continued collaboration with your esteemed institution in the future.

Thanking you.

Yours sincerely,
For R & R Apparel Sourcing

[Authorized Signatory Name]
[Designation]

STOCK LEVEL PREDICTION AND CONTROL SOFTWARE

Submitted by

PRATHEEBA S

(Reg. No.: 2428B0333)

Under the Guidance of

Dr. M. Manoj, M.C.A., M.Phil., Ph.D.

Assosiate Professor

Department of Computer Science with Data Analytics



DEPARTMENT OF COMPUTER SCIENCE WITH DATA ANALYTICS

KPR COLLEGE OF ARTS SCIENCE AND RESEARCH

(Approved by Government of Tamilnadu)

Affiliated to Bharathiar University, Coimbatore

Arasur, Coimbatore – 641407

OCTOBER 2026

STOCK LEVEL PREDICTION AND CONTROL SOFTWARE

Submitted by

PRATHEEBA S

(Reg. No.: 2428B0333)

Under the Guidance of

Dr. M. Manoj, M.C.A., M.Phil., Ph.D.

Associate Professor

Department of Computer Science with Data Analytics



DEPARTMENT OF COMPUTER SCIENCE WITH DATA ANALYTICS

KPR COLLEGE OF ARTS SCIENCE AND RESEARCH

(Approved by Government of Tamilnadu)

Affiliated to Bharathiar University, Coimbatore

Arasur, Coimbatore – 641407

OCTOBER 2026

STOCK LEVEL PREDICTION AND CONTROL SOFTWARE

Submitted by

PRATHEEBA S

(Reg. No.: 2428B0333)

Under the Guidance of

Dr. M. Manoj, M.C.A., M.Phil., Ph.D.

Associate Professor

Department of Computer Science with Data Analytics

In partial fulfillment of the requirements for the

Award of the Degree of

B.Sc. Computer Science with Data Analytics



DEPARTMENT OF COMPUTER SCIENCE WITH DATA ANALYTICS

KPR COLLEGE OF ARTS SCIENCE AND RESEARCH

(Approved by Government of Tamilnadu)

Affiliated to Bharathiar University, Coimbatore

Arasur, Coimbatore – 641407

OCTOBER 2026

KPR COLLEGE OF ARTS SCIENCE AND RESEARCH

(Approved by Government of Tamilnadu)

Affiliated to Bharathiar University, Coimbatore

Arasur, Coimbatore – 641407

CERTIFICATE

This is to certify that this project report entitled

STOCK LEVEL PREDICTION AND CONTROL SOFTWARE

is a Bonafide record of project work done by

PRATHEEBA S

(Reg.No.: 2428B0333)

Submitted in partial fulfillment of the requirements for the Degree of

B.SC. COMPUTER SCIENCE WITH DATA ANALYTICS

_____ _____

Faculty Guide Head of the Department

Submitted for Viva-Voce Examination held on _____.

_____ _____

Internal Examiner External Examiner

Office Seal

KPR COLLEGE OF ARTS SCIENCE AND RESEARCH

(Approved by Government of Tamilnadu)

Affiliated to Bharathiar University, Coimbatore

Arasur, Coimbatore – 641407

DECLARATION

I **PRATHEEBA S (Reg.No:2428B0333)** hereby declare that the project **STOCK LEVEL PREDICTION AND CONTROL SOFTWARE** submitted to **KPR COLLEGE OF ARTS SCIENCE AND RESEARCH**, partial fulfilment for the award of degree of **B.SC. COMPUTER SCIENCE WITH DATA ANALYTICS** is a record of original project work done by me during the academic year 2025-2026 under the supervision and guidance of **Dr. M. Manoj, M.C.A., M.Phil., Ph.D. Associate Professor, Department of Computer Science with Data Analytics,** KPR College of Arts Science and Research, and that it has not formed the basis for the award of any Degree/Diploma/Associate ship/fellowship or other similar title to any candidate of any University.

Signature of Student

Date:

Place: Coimbatore

ACKNOWLEDGEMENT

ACKNOWLEDGEMENT

I wish to extend my profound gratitude to **Dr. K. P. Ramasamy,** Chairman of the KPR Group, for graciously providing the institutional facilities and resources essential for the completion of this project.

I would like to express our gratitude to **Mrs. Gayathri Anandakrishnan**, Secretary of the KPR Institutions, for her visionary leadership and for fostering an academic environment that prioritizes student excellence and research.

I sincerely thank **Dr. P. Geetha**, Principal of KPR College of Arts Science and Research, for her constant encouragement and steadfast support throughout my academic tenure. I am equally grateful to **Dr. K. Pradeepa,** Dean School of Intelligent Systems and Data Science, for her valuable administrative oversight and the motivation she provided during the various stages of this project.

I am deeply indebted to **Dr. G. Satyavathy**, Head of the Department of Computer Science with Data Analytics, for her leadership and for ensuring a conducive environment that allowed me to complete my work effectively. A special word of thanks is reserved for my esteemed guide, **Dr. M. Manoj**, Associate Professor Department of Computer Science with Data Analytics, for her scholarly expertise and patient guidance, which were pivotal in shaping this study.

I also appreciate the collective support and constructive feedback provided by the entire faculty of the Department of Computer science with data Analytics, who contributed to my learning experience. Finally, I offer my humble prayers to the Almighty God for granting me the strength, sound health, and mental perseverance required to successfully conclude this project journey

PRATHEEBA S

CONTENTS

TABLE CONTENTS

CONTENTS	Pg. No.
ACKNOWLEDGEMENT	
SYNOPSIS	
1. INTRODUCTION	1
1.1 Organization Profile	3
1.2 System Specification	4
1.2.1 Hardware Configuration	6
1.2.2 Software Configuration	7
2. SYSTEM STUDY	14
2.1 Existing System	15
2.1.1 Drawbacks	15
2.2 Proposed System	17
2.2.1 Features	19
2.2.2 Modules	21
3. SYSTEM DESIGN AND DEVELOPMENT	22
3.1 Form Design	23
3.2 Input Design	24
3.3 Output Design	25
3.4 Database Design	26
3.5 System Development	27
3.5.1 Dashboard Module	28
3.5.2 Products Module	29
3.5.3 Inventory Control Module	30
3.5.4 Stock Transfer Module	30
3.5.5 Supplier Module	31

3.5.6 Purchase Order Module	32
3.5.7 POS & Billing Module	34
3.5.8 Return & Tracking Module	35
3.5.9 Barcode Scanning Module	36
3.5.10 AI Stock Prediction Module	37
3.5.11 Advanced Analysis Module	37
3.5.12 Customers CRM Module	38
3.5.13 User Authentication Module	39
3.5.14 P&L Analysis Module	40
3.5.16 Live Tracking Module	41
3.5.17 Manager Dashboard Module	42
3.5.18 Customer Management Module	43
3.5.19 Logout Module	44
4. SOFTWARE TESTING AND IMPLEMENTATION	45
4.1 Testing	45
4.2 Test cases	46
5. CONCLUSION	47
5.1. Future Enhancement	48
BIBLIOGRAPHY	52
APPENDICES	54
A. Data Flow Diagram	54
B. Table Structure	56
C. Sample Coding	62
D. Sample Input	72
E. Sample Output	73

SYNOPSIS

SYNOPSIS

The Stock Level Prediction and Control Software is an enterprise-level inventory management system developed under the StockSense Enterprise concept for R&R Apparel Sourcing. The system is designed to efficiently manage and monitor products, inventory, suppliers, warehouses, purchase orders, stock transfers, sales, returns, and business reports through a centralized web-based platform. It addresses common problems in traditional inventory management, such as manual stock calculations, delayed updates, stock shortages, overstocking, difficulty in tracking inventory movements, and limited analytical support. The system provides an interactive dashboard with important Key Performance Indicators (KPIs), sales trends, inventory health, stock movement, supplier performance, and category-wise analytics, enabling managers to understand the overall condition of the organization's inventory quickly.

A major objective of the system is to maintain optimum stock levels and support better inventory decisions using Data Analytics and AI-based Demand Prediction. The system analyzes historical demand and current inventory information to predict future product demand and provide recommended reorder quantities. It displays information such as current stock, average demand, predicted demand, reorder point, recommended order quantity, forecast accuracy, demand trend, and confidence score. The Automated Reorder feature helps identify products that require replenishment and classifies recommendations according to priority. The system also provides Inventory Health categories such as Healthy Stock, Low Stock, Critical Stock, Overstock, and Out of Stock.

The application contains several functional modules including Product Management, Inventory Control, Warehouse Management, Supplier Management, Purchase Order Management, Stock Transfer, POS and Billing, Returns and Tracking, Reports, Authentication, Barcode/QR Scanning, Excel Import/Export, and StockSense AI Chatbot. It supports multiple warehouses and maintains realistic information about products, suppliers, transactions, purchases, transfers, and returns. Search, filtering, sorting, and pagination features allow users to efficiently work with large datasets. The system also maintains data consistency between stock, sales, purchase, and analytical information so that dashboard values and reports are generated from the underlying dataset.

Overall, the Stock Level Prediction and Control Software provides the system helps organizations monitor stock levels, analyze business performance, identify inventory problems, manage suppliers and warehouses, track purchases and sales, predict future demand, and make informed replenishment decisions.

1. INTRODUCTION

Inventory management is an essential part of every organization that purchases, stores, and sells products. Maintaining the right quantity of stock is important because insufficient inventory can lead to stock shortages, delayed customer orders, and loss of sales, while excessive inventory can increase storage costs and result in overstocking. As the number of products, suppliers, warehouses, sales transactions, and purchase orders increases, traditional manual methods of managing inventory become difficult, time-consuming, and prone to errors. Therefore, organizations require an efficient and intelligent system that can monitor inventory continuously and support timely decision making.

The Stock Level Prediction and Control Software is developed as an enterprise-oriented inventory management solution under the StockSense Enterprise concept for R&R Apparel Sourcing. The system provides a centralized platform for managing products, inventory, suppliers, warehouses, purchase orders, stock transfers, POS and billing transactions, returns, reports, and analytics. It is designed to replace fragmented manual processes with an integrated web-based system that provides better visibility into stock and business operations.

The main purpose of the proposed system is to improve stock visibility and help organizations maintain appropriate inventory levels. Instead of treating inventory as a simple list of products, the system connects stock information with sales, purchasing, warehouse movement, supplier performance, and historical demand. This integrated approach allows users to understand not only how much stock is currently available, but also how stock is moving and whether future replenishment may be required.

One of the major components of the system is the enterprise-level Dashboard Summary. The dashboard provides a consolidated view of important operational metrics through KPI cards and analytical visualizations. The proposed dashboard includes values such as Total Products, Total Stock Units, Inventory Value, Low Stock Items, Out of Stock, Overstock Items, Pending Purchase Orders, Active Suppliers, Today's Sales, Monthly Sales, Stock Turnover Rate, and Forecast Accuracy. These indicators help management obtain a quick overview of inventory and business performance without manually checking multiple pages or records.

The dashboard also provides meaningful charts for analyzing inventory and sales information. These include Inventory Stock Trend, Sales Trend, Category Distribution, Low Stock Analysis, Top Selling Products, Inventory Value by Category, Supplier Performance, AI Demand Forecast, and Stock Movement. The purpose of these visualizations is to convert large amounts of operational data into

information that can be understood quickly. For example, an inventory trend can show stock received, stock sold, and current inventory over a period of time, while a sales trend can help users identify changes in revenue and units sold. Supplier performance can be compared using delivery rate, order fulfillment, quality score, and purchase value.

The proposed system also introduces an Inventory Health section to provide a direct understanding of stock conditions. Products can be classified as Healthy Stock, Low Stock, Critical Stock, Overstock, or Out of Stock. Progress indicators and percentages can be used to show the overall distribution of inventory health. This helps managers identify potential problems at an early stage and focus attention on products that require corrective action.

Product management is another important part of the system. The Products Catalog is designed to maintain a large and realistic dataset containing information such as SKU ID, product name, brand, category, subcategory, supplier, warehouse, cost price, selling price, current stock, reserved stock, available stock, reorder level, maximum stock, units sold, monthly demand, lead time, stock status, profit margin, last restocked date, last sold date, and product rating. The application supports multiple categories, including Electronics, Computer Accessories, Furniture, Office Supplies, Apparel, Footwear, Home Appliances, Mobile Accessories, Networking, Storage Devices, Personal Care, and Sports & Fitness.

The POS and Billing module connects sales activity with inventory management. Sales transactions contain information such as Invoice ID, Date, Customer, Product, SKU, Quantity, Unit Price, Discount, Tax, Total Amount, Payment Method, Store, and Status. The system supports payment methods including UPI, Card, Cash, and Net Banking. By connecting sales transactions with stock information, the system can provide better visibility into products sold, revenue generated, and changes in inventory levels. Similarly, the Returns and Tracking module records returned products and associated refund information. It stores Return ID, Invoice ID, Product, Customer, Quantity, Return Reason, Return Date, Refund Amount, and Status. Return reasons may include Damaged, Wrong Product, Customer Request, Defective, and Size Issue.

The proposed system also retains a range of supporting features, including Authentication, Owner Session, StockSense AI chatbot, AI Demand Prediction, Barcode/QR Scanning, Excel Import/Export, Reports, Dashboard, Products, Inventory Control, Stock Transfers, Suppliers, Purchase Orders, POS and Billing, Returns and Tracking, and Light Mode. These features form an integrated platform rather than a collection of independent functions.

1.1 ORGANIZATION PROFILE

The organization considered for this project is a business environment that manages a wide range of products through an integrated inventory and stock control process. The organization is involved in purchasing, storing, monitoring, selling, and distributing products through multiple operational activities. As the volume of products, suppliers, customers, sales transactions, and warehouse operations increases, effective inventory management becomes essential for maintaining smooth and profitable business operations.

Inventory is one of the major resources of an organization because it directly affects sales, customer satisfaction, purchasing decisions, warehouse utilization, and overall profitability. Maintaining an appropriate stock level is therefore important to ensure that products are available whenever they are required. Excess inventory can increase storage and holding costs, while insufficient inventory can lead to stock shortages and loss of potential sales.

The organization requires a centralized system to maintain product details, stock quantities, supplier information, warehouse records, purchase orders, sales transactions, stock transfers, and return information. The proposed Stock Level Prediction and Control Software is developed to fulfill this requirement by bringing these activities together in a single web-based platform. The system enables authorized users to monitor inventory information, perform stock-related operations, and obtain analytical insights from centralized data.

The organization may operate through multiple warehouses located in different regions. Each warehouse can maintain information regarding total products, total stock units, inventory value, low-stock items, capacity utilization, and stock movement. The system also supports movement of products between warehouses through the stock transfer module, allowing users to monitor source warehouse, destination warehouse, quantity, transfer date, expected arrival, and transfer status.

Supplier management is another important organizational activity. The system maintains supplier details such as supplier ID, supplier name, contact information, location, products supplied, total orders, pending orders, on-time delivery percentage, quality score, total purchase value, and supplier status. This information helps users evaluate supplier performance and support better purchasing decisions.

The organization also handles sales and customer transactions. The POS and Billing module records invoice details, customers, products, quantities, prices, discounts, taxes, payment methods, stores, and transaction status. The Returns and Tracking module maintains information about returned products,

return reasons, refund amounts, dates, and statuses. These features provide a connected view of product movement from purchasing and storage to sales and returns.

To improve inventory decision-making, the organization can use the system's AI Demand Prediction and Automated Reorder features. Historical demand and current stock information are analyzed to estimate future requirements and provide reorder recommendations. This allows users to identify products that may require replenishment and take action before inventory reaches critical levels.

Overall, the organization provides a suitable environment for implementing the Stock Level Prediction and Control Software. The proposed system integrates inventory management, purchasing, warehouse operations, sales, returns, data analytics, visualization, and AI-based prediction into one centralized platform. It is designed to improve inventory visibility, reduce manual effort, support timely replenishment, and provide better decision-making capabilities.

1.2 SYSTEM SPECIFICATION

The Stock Level Prediction and Control Software is a web-based inventory management system developed to provide a centralized platform for managing and monitoring stock-related activities. The system is designed to maintain product information, inventory levels, suppliers, warehouses, purchase orders, stock transfers, sales transactions, returns, and reports in an organized manner. The main purpose of the system is to reduce the difficulties associated with manual inventory management and provide users with accurate and easily accessible information. By integrating different inventory operations into one platform, the system helps users monitor stock efficiently and supports better decision making.

The system provides an interactive Dashboard Summary that presents important information about the overall condition of inventory and business operations. The dashboard includes Key Performance Indicators such as Total Products, Total Stock Units, Inventory Value, Low Stock Items, Out of Stock Items, Overstock Items, Pending Purchase Orders, Active Suppliers, Today's Sales, Monthly Sales, Stock Turnover Rate, and Forecast Accuracy. These metrics allow users to understand the current inventory position quickly and identify areas that require attention. The dashboard also includes trend indicators and previous-period comparisons to make the presented information more meaningful.

The system includes several analytical charts for representing inventory and business information visually. The Inventory Stock Trend displays stock received, stock sold, and current inventory over a selected period. The Sales Trend presents sales performance and revenue information, while Category

Distribution shows the distribution of products across different categories. Low Stock Analysis helps users identify products approaching their reorder level, and Top Selling Products provides information about products with high sales volume. Inventory Value by Category can be used to compare the financial value of inventory across categories. Supplier Performance can also be analyzed using delivery rate, order fulfillment, quality score, and purchase value. In addition, the AI Demand Forecast presents historical demand, predicted demand, confidence range, and future demand forecasts.

The Inventory Health section is designed to provide a clear understanding of stock conditions. Products can be classified into Healthy Stock, Low Stock, Critical Stock, Overstock, and Out of Stock categories. Progress indicators and percentages can be used to represent these conditions visually. This allows users to identify inventory problems quickly and take appropriate action. The system is designed so that these values are dynamically connected to the dataset, ensuring that the information displayed reflects the actual inventory condition.

The Product Catalog is another major component of the system. Each product can contain information such as SKU ID, product name, brand, category, subcategory, supplier, warehouse, cost price, selling price, current stock, reserved stock, available stock, reorder level, maximum stock, units sold, monthly demand, lead time, stock status, profit margin, last restocked date, last sold date, and product rating. The system is designed to support a larger product dataset with multiple categories. This provides sufficient information for inventory analysis, sales analysis, demand forecasting, and stock control.

The system also supports Multiple Warehouse Management. Different warehouses can be maintained with information about total products, total stock units, inventory value, low-stock count, capacity utilization, and stock movement. Stock can be transferred from one warehouse to another through the Stock Transfer module. Transfer records include the product, SKU, source warehouse, destination warehouse, quantity, transfer date, expected arrival, and status. This feature provides better control over inventory stored at different locations.

Supplier Management enables the system to maintain supplier details and performance information. Supplier records can include Supplier ID, Supplier Name, Contact, Location, Products Supplied, Total Orders, Pending Orders, On-Time Delivery Percentage, Quality Score, Total Purchase Value, and Supplier Status. This information helps users monitor supplier performance and support purchasing decisions. The Purchase Order Module maintains purchase order number, supplier, warehouse, order date, expected delivery, products, quantity, total value, payment status, delivery status, and priority.

Purchase orders can be tracked through statuses such as Pending, Approved, Processing, Shipped, Delivered, Delayed, and Cancelled.

The system provides Search**,** Filter**,** Sort**,** and Pagination features to improve usability when dealing with large datasets. Users can search for products by SKU, product name, supplier, or category. They can filter records according to warehouse, stock status, supplier, and date, and sort records by stock, price, sales, or inventory value. Pagination prevents a very large number of records from being displayed on a single page and makes the system easier to use.

The system also maintains important supporting functionality such as Authentication, Owner Session, StockSense AI chatbot, AI Demand Prediction, Barcode/QR Scanning, Excel Import/Export, Reports, Dashboard, Products, Inventory Control, Stock Transfers, Suppliers, Purchase Orders, POS and Billing, Returns and Tracking, and Light Mode. These features work together to form a complete inventory management environment.

1.2.1 HARDWARE CONFIGURATION

The Stock Level Prediction and Control Software is a web-based application that can be developed and operated using a standard computer system. The hardware requirements are moderate because the application mainly performs inventory management, database operations, data visualization, report generation, and AI-based demand prediction. The system should have sufficient processing power, memory, storage, and network connectivity to ensure smooth operation.

A multi-core processor such as Intel Core i3/i5 or an equivalent AMD processor is suitable for developing and running the application. A minimum of 8 GB RAM is recommended so that the web application, database, development tools, browser, and analytics components can run efficiently at the same time. Additional RAM can be used when handling larger datasets or running prediction and reporting operations.

The system should have at least 256 GB of storage, preferably using an SSD, to store the application files, database, datasets, reports, and other project resources. A full HD monitor with a resolution of 1920 × 1080 is recommended for displaying the dashboard, tables, charts, and other interfaces clearly. A standard keyboard and mouse are sufficient for user interaction.

A stable internet or local network connection may be required for accessing the web application, downloading dependencies, updating resources, and supporting future online deployment. The system

can initially be operated on a local computer for development and demonstration and can later be deployed on a suitable server for multiple users.

Hardware Component	Specification
Processor	Intel Core i3/i5 or equivalent AMD processor
RAM	Minimum 8 GB
Storage	Minimum 256 GB SSD
Monitor	Full HD (1920 × 1080)
Keyboard	Standard USB/Wireless Keyboard
Mouse	Standard USB/Wireless Mouse

Table 1.2.1 Hardware Configuration

1.2.2 SOFTWARE CONFIGURATION

The Stock Level Prediction and Control Software is designed as a web-based application that uses different software technologies for application development, database management, data analytics, visualization, and user interaction. The software environment provides the necessary support for developing and operating the system efficiently.

The front end of the application can be developed using HTML, CSS, JavaScript, and Bootstrap. HTML is used to create the structure of web pages, while CSS is used to design the layout, colors, fonts, tables, forms, and dashboard components. JavaScript is used to provide interactive functionality such as search, filtering, validation, dynamic updates, and dashboard interactions. Bootstrap can be used to create responsive layouts that work properly on desktops, tablets, and mobile devices.

The backend of the system is developed using Python with the Flask framework. Flask handles application routing, user requests, authentication, session management, database communication, and business logic. It also connects the different modules of the system such as products, inventory, suppliers, purchase orders, sales, returns, and reports.

For data visualization, charts and graphical representations can be created using suitable JavaScript or Python visualization libraries. These visualizations are used to display sales trends, inventory trends, category distribution, supplier performance, stock movement, and demand forecasts in an understandable format.

Software Component	Recommended Technology
Operating System	Windows / Linux
Front End	HTML, CSS, JavaScript
UI Framework	Bootstrap
Backend	Python
Web Framework	Flask
Database	SQLite / MySQL
Data Processing	Pandas
Machine Learning	Scikit-learn
Visualization	JavaScript/Python Chart Libraries
Browser	Google Chrome / Microsoft Edge

Table 1.2.2 Software Configuration

Operating System – Windows 11

Windows 11 is used as the operating system for developing and running the Production System. It provides a stable and user-friendly environment that supports various development tools and applications. The operating system plays a key role in managing system resources such as memory, processor, and storage.

Windows 11 offers compatibility with modern software tools like Visual Studio Code, Python, and web browsers, making it suitable for web application development. It also provides security features, multitasking capabilities, and efficient file management, which help in smooth development and execution of the system.

Programming Language – Python

Python is the primary programming language used in the development of the Production System. It is a high-level, interpreted programming language known for its simplicity, readability, and versatility. Python allows developers to write clear and structured code, making it easier to understand, maintain, and modify the application over time.

One of the major advantages of Python is its easy-to-learn syntax, which resembles natural language. This makes it suitable for both beginners and experienced developers. The simplicity of Python reduces development time and helps in building applications quickly without compromising on functionality.

Python supports multiple programming paradigms, including procedural, object-oriented, and functional programming. This flexibility allows developers to choose the most suitable approach for implementing different parts of the system. In the Production System, Python is used to handle backend logic such as processing user inputs, managing production data, and performing operations related to workers, looms, and reports.

Another important feature of Python is its extensive library support. Python provides a wide range of built-in and external libraries that simplify complex tasks such as database operations, data processing, and web development. These libraries reduce the need for writing repetitive code and improve overall development efficiency.

Python is also platform-independent, meaning that applications developed using Python can run on different operating systems such as Windows, Linux, and macOS without significant modifications. This makes the Production System more flexible and adaptable to different environments.

In addition, Python integrates easily with web frameworks such as Flask, which is used in this project to develop the backend of the system. This integration allows smooth communication between the user interface and the database, ensuring efficient data processing and system performance.

Python also supports rapid application development, enabling developers to build prototypes and implement features quickly. This is particularly useful in project development where time and efficiency are important factors.

Furthermore, Python has strong community support and extensive documentation, which makes it easier to troubleshoot issues and learn new concepts. Developers can access a large number of online resources, tutorials, and forums to enhance their knowledge and improve the system.

Framework – Flask

Flask is a lightweight web framework used for developing the backend of the Production System. It provides the necessary tools to build web applications, handle HTTP requests, and manage routing between different web pages.

Flask allows developers to create scalable and flexible applications without unnecessary complexity. It supports integration with databases, template engines, and external libraries. In this project, Flask is used to process user input, handle business logic, and communicate with the database, ensuring smooth system operation.

Frontend Technologies – HTML, CSS, JavaScript

The frontend of the Production System is developed using HTML, CSS, and JavaScript, which together create an interactive and user-friendly interface.

- **HTML (HyperText Markup Language)** is used to define the structure of web pages, including forms, tables, and navigation elements.
- **CSS (Cascading Style Sheets)** is used to design and style the user interface, improving the visual appearance and layout of the system.
- **JavaScript** is used to add interactivity and dynamic behavior, such as form validation, real-time updates, and user interaction handling.

These technologies ensure that the system interface is responsive, easy to use, and accessible across different devices.

Database – SQLite

SQLite is used as the database management system for the Production System. It is a lightweight, serverless, and self-contained database engine that is widely used for small to medium-scale applications. SQLite does not require a separate server to operate, which makes it simple to set up and easy to integrate with applications developed using Python.

One of the major advantages of SQLite is its simplicity and ease of use. Unlike other database systems, SQLite stores the entire database in a single file, which reduces the complexity of database management. This makes it highly suitable for projects where quick setup and minimal configuration are required.

In the Production System, SQLite is used to store all production-related data such as worker details, loom information, daily production records, and salary details. The database is structured into multiple tables, each designed to store specific types of data. For example, separate tables are maintained for workers, looms, and daily production entries, ensuring that the data is organized and easy to manage.

SQLite supports Structured Query Language (SQL), which allows efficient execution of database operations such as inserting, updating, deleting, and retrieving data. SQL queries are used within the application to interact with the database and perform necessary operations. This enables the system to handle production data efficiently and respond quickly to user requests. Another important feature of SQLite is its fast performance and efficient data handling. Since it operates directly on a file without the

need for a database server, it provides quick data access and reduced latency. This ensures that the Production System can retrieve and display data instantly, improving the overall user experience.

SQLite also ensures data integrity and consistency through the use of constraints such as primary keys and data types. These constraints help maintain the accuracy and reliability of stored data. For example, unique identifiers such as worker IDs and production IDs ensure that each record is distinct and properly managed.

In addition, SQLite is highly reliable and portable. The database file can be easily transferred between systems without compatibility issues, making it convenient for backup and deployment. It also supports transaction management, which ensures that database operations are completed successfully without data loss.

SQLite integrates seamlessly with Python using built-in libraries, which simplifies database connectivity and operations. This integration allows developers to perform database actions directly from the application code, ensuring smooth interaction between the system and the database.

Overall, SQLite provides an efficient, reliable, and easy-to-manage solution for handling data in the Production System. Its lightweight nature, fast performance, and strong integration with Python make it an ideal choice for developing and maintaining the system database.

IDE / Code Editor – Visual Studio Code

Visual Studio Code (VS Code) is used as the primary Integrated Development Environment (IDE) for developing the Production System. It is a lightweight, powerful, and widely used code editor developed by Microsoft. VS Code provides a flexible development environment that supports multiple programming languages, including Python, HTML, CSS, and JavaScript, making it highly suitable for full-stack web development.

One of the key advantages of Visual Studio Code is its user-friendly interface, which allows developers to easily write, edit, and manage code. The editor provides features such as syntax highlighting, auto-completion, and code suggestions, which help in writing accurate and efficient code. These features reduce coding errors and improve development speed.

VS Code also supports a wide range of extensions and plugins that enhance its functionality. For this project, extensions such as Python support, code formatters, and debugging tools are used to simplify development. These extensions allow seamless integration with Flask applications and improve

productivity by automating repetitive tasks. Another important feature of Visual Studio Code is its built-in debugging capability. Developers can identify and fix errors in the code by setting breakpoints, inspecting variables, and monitoring program execution. This helps in improving code quality and ensuring that the system functions correctly.

VS Code also provides an integrated terminal, which allows developers to run commands, execute Python scripts, and manage project files directly within the editor. This eliminates the need to switch between different tools and improves workflow efficiency.

The editor supports version control integration, especially with Git, which helps developers track changes in the code and manage different versions of the project. This is useful for maintaining project history and ensuring proper code management.

Additionally, Visual Studio Code is highly customizable, allowing developers to modify themes, layouts, and settings according to their preferences. This flexibility creates a comfortable development environment and enhances overall productivity.

Furthermore, VS Code provides fast performance and requires minimal system resources compared to traditional IDEs. This makes it suitable for systems with moderate hardware configurations while still delivering powerful development features.

Overall, Visual Studio Code plays a significant role in the development of the Production System by providing an efficient, flexible, and feature-rich environment. Its advanced tools and extensions help streamline the development process, improve code quality, and ensure smooth implementation of the system.

Machine Learning Library – Scikit-learn

Scikit-learn is an open-source Machine Learning library in Python that provides efficient tools for data analysis and predictive modeling. It is used to implement intelligent features that assist administrators in monitoring production and making better decisions. The library is utilized for the Isolation Forest algorithm to detect abnormal loom production patterns and the Linear Regression algorithm to forecast workers' weekly production based on their current performance. These predictive capabilities help identify potential machine issues and estimate whether workers are likely to achieve production targets.

Data Analysis Libraries – Pandas and NumPy

Pandas and NumPy are powerful Python libraries used for data processing and numerical computation. Pandas provides flexible data structures for organizing, filtering, and analyzing production records, while NumPy supports high-performance mathematical and numerical operations. In this project, these libraries are used to process daily production data, prepare datasets for machine learning models, calculate statistical values, and perform efficient data manipulation required for production analytics and forecasting.

Visualization Library – Chart.js

Chart.js is a JavaScript library used to create interactive and responsive graphical charts. It enables the system to display production trends, worker performance, salary summaries, and other analytical information in an easy-to-understand graphical format. The library enhances data visualization by presenting information through bar charts, line charts, and other graphical representations, allowing administrators to analyze production performance and make informed decisions more effectively.

Web Browser – Google Chrome

Google Chrome is used as the web browser for testing and running the Production System. It provides fast performance and supports modern web standards, ensuring compatibility with the application. Chrome includes built-in developer tools that help in debugging, inspecting elements, and analyzing application performance. These tools allow developers to identify and fix issues quickly, ensuring that the system functions correctly and efficiently.

This software configuration provides a stable and efficient platform for developing and running the Production System. The combination of these technologies ensures that the application is scalable, easy to maintain, and capable of supporting future enhancements.

It provides fast performance and supports modern web standards, ensuring compatibility with the application. Chrome includes built-in developer tools that help in debugging, inspecting elements, and analyzing application performance. These tools allow developers to identify and fix issues quickly.

This platform is used for developing and running the Production System. The combination of these technologies ensures that the application is scalable, easy to maintain, and capable of supporting future enhancements.

2. SYSTEM STUDY

The System Study is carried out to understand the existing inventory management process and to determine the requirements for developing the proposed Stock Level Prediction and Control Software. The study examines how products, stock, suppliers, warehouses, purchasing, sales, transfers, and returns are currently handled and identifies the problems that may occur when these activities are managed manually or through disconnected systems. The purpose of the study is to provide a clear foundation for designing a centralized inventory management and analytics platform.

Inventory management involves continuous monitoring of products and stock quantities. Every organization needs to know how much stock is available, which products are selling quickly, which products require replenishment, and which products are remaining unused for long periods. When inventory information is not maintained properly, the organization may face stock shortages, overstocking, inaccurate records, delayed purchasing, and difficulty in responding to customer requirements. Therefore, a systematic study of the existing process is required before developing an improved system.

The existing environment may involve maintaining product details, stock information, purchase records, supplier details, sales records, and return information through separate files or manual records. Information may be updated whenever a product is received, sold, transferred, or returned. However, as the number of products and transactions increases, maintaining this information manually becomes more difficult. The proposed system addresses these issues by connecting the different activities through a centralized application.

The system study also considers the need for analytical support. A modern inventory system should not only record transactions but should also help users understand what the stored data means. Sales history, stock movements, supplier performance, inventory value, and product demand can provide useful information for decision making. The proposed system therefore includes dashboard analytics, inventory health monitoring, demand forecasting, and reorder recommendations as part of the overall inventory control process.

The organization may face stock shortages, overstocking, inaccurate records, delayed purchasing, and difficulty in responding to customer requirements. Therefore, a systematic study of the existing process is required before developing an improved system.

2.1 EXISTING SYSTEM

The existing inventory management system generally relies on manual records, spreadsheets, or separate files for maintaining product and stock information. Product details, supplier information,

purchase records, sales transactions, warehouse details, and return information may be maintained independently. Whenever a product is purchased, sold, transferred, or returned, the corresponding stock information needs to be updated manually.

This method can be manageable when the number of products and transactions is small. However, as the inventory grows, maintaining accurate and updated information becomes difficult. Users may need to check multiple records to determine the actual stock availability of a product. Manual data entry can also result in incorrect quantities, duplicate records, missing updates, and inconsistencies between actual stock and recorded stock.

The existing system also provides limited support for analyzing inventory data. Managers may have to manually calculate sales trends, inventory value, stock movement, and product demand before making decisions. Identifying low-stock, critical-stock, or overstock products can also require continuous manual monitoring.

Another limitation is the lack of intelligent demand forecasting. Traditional inventory methods generally depend on fixed reorder levels or manual judgement and do not effectively use historical sales information to estimate future demand. As a result, purchasing decisions may not always match actual customer requirements.

Therefore, the existing system has limitations in centralized data management, real-time monitoring, analytical reporting, demand prediction, and automated stock control, creating a need for a more efficient and intelligent inventory management solution.

2.1.1. DRAWBACKS

The major drawbacks of the existing system are:

- **Manual Data Entry:** The existing system depends heavily on manual entry of product, stock, purchase, and sales information. This increases the workload of employees and creates a greater possibility of entering incorrect, duplicate, or incomplete data. Maintaining records manually also becomes difficult when the number of products and transactions increases.

- **Inaccurate Stock Information:** Stock quantities may not always be updated immediately after products are purchased, sold, transferred, or returned. Because of delayed updates, the recorded stock may differ from the actual stock available in the warehouse. This can create problems in inventory planning and order processing.
- **Time-Consuming Inventory Management:** Users may need to check several records or files to find product details, stock quantities, supplier information, or transaction history. This increases the amount of time required for routine inventory activities and reduces the overall efficiency of the organization.
- **Lack of Centralized Information:** Product details, supplier records, purchase information, sales transactions, and warehouse details may be maintained separately. The absence of a centralized system makes it difficult to obtain a complete view of inventory operations and may result in duplicated or inconsistent information.
- **Difficulty in Identifying Low Stock:** In a large inventory, manually checking every product to determine whether it has reached its reorder level is difficult. Products may become critically low before management notices the situation, increasing the possibility of stock shortages and interrupted sales.
- **Risk of Overstocking:** The existing system may not provide sufficient information about products that are being stored in excessive quantities. Without proper demand analysis, organizations may continue purchasing products even when the available stock is already higher than the required level, increasing storage and holding costs.
- **Lack of Demand Prediction:** Traditional inventory systems generally focus on current stock levels and fixed reorder points. They may not analyze historical sales data to estimate future demand, making it difficult for managers to predict which products will be required in the coming days or months.
- **Manual Reordering Process:** Replenishment decisions are often based on manual checking and personal judgement. Managers must identify low-stock products, estimate the required quantity, and contact suppliers separately. This process can cause delays and may result in ordering too much or too little stock.
- **Limited Analytical Support:** The existing system may provide basic inventory records without meaningful charts, trends, or analytical summaries. Managers may therefore need to calculate sales performance, inventory value, stock movement, and supplier performance manually before making important decisions.

- **Difficulty in Multi-Warehouse Monitoring:** Managing inventory across multiple warehouses becomes complicated when each location maintains separate records. Tracking available stock, transfers, and warehouse capacity requires additional effort and makes it difficult to obtain a complete organization-wide inventory view.
- **Poor Decision-Making Support:** Since the existing approach provides limited real-time analytics and prediction capabilities, managers may have to depend mainly on manual calculations and experience. This can make inventory decisions slower and less effective.

2.2 PROPOSED SYSTEM

The proposed Stock Level Prediction and Control Software is designed to overcome the limitations of the existing inventory management approach by providing a centralized, efficient, and intelligent platform for managing inventory and related business activities. The system integrates product management, inventory control, supplier management, warehouse management, purchase orders, stock transfers, POS and billing, returns, reporting, and analytics into a single application. This centralized approach reduces the need for maintaining separate records and makes inventory information easier to access and manage.

The proposed system provides real-time stock monitoring by maintaining current stock, reserved stock, available stock, reorder level, and maximum stock information. It helps users identify healthy, low, critical, overstock, and out-of-stock products and take appropriate action. A professional dashboard provides important KPIs such as total products, inventory value, low-stock items, out-of-stock products, sales, purchase orders, suppliers, stock turnover, and forecast accuracy. The dashboard also includes meaningful charts for sales trends, inventory trends, category distribution, top-selling products, supplier performance, stock movement, and inventory value, helping managers understand business performance more effectively.

One of the major advantages of the proposed system is the integration of AI-based Demand Prediction. The system uses historical demand information to estimate future product requirements and provides details such as current stock, historical average demand, predicted demand, reorder point, recommended order quantity, forecast accuracy, demand trend, and confidence score. The system can also provide forecasts for future periods and display historical and predicted demand through charts. This helps managers make more informed purchasing and inventory planning decisions.

The system provides a professional Dashboard Summary that presents important inventory and business information through KPI cards and analytical charts. The dashboard can display total products, total stock units, inventory value, low-stock items, out-of-stock items, overstock items, pending purchase orders, active suppliers, sales, stock turnover rate, and forecast accuracy. It also provides charts for inventory trends, sales trends, category distribution, low-stock analysis, top-selling products, supplier performance, stock movement, and inventory value. These features help users understand the current condition of the organization and support better decision making.

The proposed system also includes an Automated Reorder feature that provides recommendations when products require replenishment. Instead of depending completely on manual stock checking, the system can use current stock and predicted demand to suggest a suitable reorder quantity. Recommendations can be classified according to priority, such as Critical, High, Medium, and Low, helping users focus on the most important inventory requirements. This helps reduce manual work, improve the accuracy of inventory information, and provide users with quick access to important stock details.

A major feature of the proposed system is AI Demand Prediction and Automated Reorder. Historical demand information is analyzed to estimate future product requirements. For selected products, the system can display current stock, historical average demand, predicted demand, reorder point, recommended order quantity, forecast accuracy, demand trend, and confidence score. Based on this information, the automated reorder feature recommends products that need replenishment and assigns suitable priorities such as Critical, High, Medium, and Low. This helps managers take action before products become unavailable and improves inventory planning.

The proposed system also supports multiple warehouses, supplier management, purchase orders, stock transfers, POS and billing, returns tracking, barcode/QR scanning, Excel import and export, authentication, reports, and StockSense AI chatbot functionality. Search, filtering, sorting, and pagination make large datasets easier to manage. The system maintains consistent inventory calculations and uses the same underlying dataset for dashboards, charts, tables, and reports. Overall, the proposed system combines inventory management, data analytics, data visualization, and artificial intelligence to provide a professional and efficient solution for stock control and data-driven decision making.

In addition, the system supports multiple warehouses, supplier performance monitoring, purchase order tracking, stock transfers, POS and billing, returns management, barcode and QR scanning, Excel

import and export, authentication, reports, and StockSense AI chatbot functionality. Search, filtering, sorting, and pagination make it easier to work with large amounts of product and transaction data.

Overall, the proposed system provides a complete inventory intelligence solution that combines inventory management, database management, data analytics, data visualization, and artificial intelligence. It helps reduce manual effort, improve stock visibility, identify inventory problems at an early stage, support better purchasing decisions, and improve overall inventory control.

2.2.1 FEATURES

(i) Product Management

- The system maintains complete information about all products in the inventory.
- It stores details such as SKU, product name, category, supplier, price, and demand.
- Users can add, update, view, and manage product records easily.
- This helps maintain an organized and accurate product catalog.

(ii) Inventory Management

- The system continuously manages current, reserved, and available stock quantities.
- It maintains reorder levels and maximum stock levels for each product.
- Stock information can be updated whenever products are purchased, sold, transferred, or returned.
- This helps maintain accurate inventory records and better stock control.

(iii) Inventory Health Monitoring

- The system classifies products according to their current stock condition.
- Products can be identified as Healthy Stock, Low Stock, Critical Stock, Overstock, or Out of Stock.
- This helps users quickly identify products that require immediate attention.
- It supports better monitoring and efficient inventory planning.

(iv) AI Demand Prediction

- The system uses historical demand information to estimate future product requirements.
- It displays predicted demand along with historical demand and demand trends.

- Forecast accuracy and confidence information can also be presented to users.
- This helps managers make better purchasing and inventory planning decisions.

(v) Automated Reorder Recommendation

- The system identifies products that may require replenishment based on stock and demand conditions.
- It provides recommended reorder quantities for selected products.
- Reorder priorities can be classified as Critical, High, Medium, or Low.
- This helps reduce stock shortages and supports timely purchasing decisions.

(vi) Enterprise Dashboard

- The system provides a centralized dashboard for monitoring inventory and business performance.
- It displays important KPIs such as products, stock, inventory value, sales, and purchase orders.
- Users can view important operational information without opening every individual module.
- This provides a quick and clear overview of the complete inventory system.

(vii) Data Visualization

- The system presents inventory and business information through meaningful charts and graphs.
- It includes sales trends, inventory trends, category distribution, stock movement, and supplier analysis.
- Charts make large amounts of data easier to understand and compare.
- This supports faster and more effective data-driven decision making.

(viii) Multi Warehouse Management

- The system supports inventory management across multiple warehouses and locations.
- It maintains warehouse information such as stock units, inventory value, and low-stock items.
- Users can monitor inventory available at different locations.
- This improves stock distribution and warehouse-level inventory control.

(ix) Supplier Management

- The system maintains complete supplier information in a centralized database.
- It stores supplier details, products supplied, orders, delivery performance, and quality scores.
- Supplier purchase value and supplier status can also be monitored.
- This helps users evaluate suppliers and make better procurement decisions.

2.2.2 MODULES

(i) User Authentication Module

Manages user login, logout, registration, sessions, and role-based access.

(ii) Product Management Module

Maintains product details such as SKU, name, category, supplier, price, stock, and demand information.

(iii) Inventory Control Module

Monitors current stock, available stock, reorder levels, and identifies low, critical, overstock, and out-of-stock products.

(iv) AI Demand Prediction Module

Analyzes historical demand data to predict future product requirements and provide forecast information.

(v) Automated Reorder Module

Generates reorder recommendations based on current stock, predicted demand, and reorder levels.

(vi) Supplier and Purchase Management Module

Manages supplier details, purchase orders, delivery status, payment information, and supplier performance.

(vii) Warehouse and Stock Transfer Module

Manages warehouse inventory and tracks the movement of products between different warehouse locations.

3. SYSTEM DESIGN

The System Design defines the overall structure and working arrangement of the Stock Level Prediction and Control Software. The system is designed as an integrated web-based application in which inventory operations, transaction processing, analytics, and prediction functions work together through a centralized database. The main purpose of the design is to provide a clear structure for how users interact with the application, how information is processed, and how data is stored and retrieved.

The system follows a modular design approach. The application is divided into eight major modules: User Authentication, Product Management, Inventory Control, AI Demand Prediction, Automated Reorder, Supplier and Purchase Management, Warehouse and Stock Transfer, and Sales, Billing and Returns. Each module performs a specific function while communicating with the common database and application services.

The presentation layer is responsible for providing the user interface of the application. It contains login pages, dashboards, forms, tables, navigation menus, charts, search controls, and report screens. Users can enter information through structured forms and view processed information through dashboards and tables.

The application handles authentication, product operations, stock calculations, purchase processing, sales transactions, stock transfers, returns, report generation, and other business rules. When a user performs an operation, the application validates the input, applies the appropriate rules, communicates with the database, and returns the required result to the interface.

The system also includes an analytics and prediction layer. This component processes inventory and sales information to generate meaningful analytical results. Dashboard indicators and charts are generated from the underlying data, while the AI Demand Prediction module uses historical demand information to estimate future requirements. The prediction section can display current stock, historical average demand, predicted demand, reorder point, recommended order quantity, forecast accuracy, demand trend, and confidence score.

The system design provides a structured framework for integrating inventory management, transactions, analytics, and artificial intelligence into one application. The combination of modular architecture, centralized data storage, dashboard visualization, predictive analysis, and automated inventory support makes the system suitable for managing large amounts of inventory information while providing users with an efficient and organized working environment.

3.1 FILE DESIGN

The File Design of the Stock Level Prediction and Control Software is developed to provide a simple, clear, and user-friendly method for entering, updating, and managing information. Each form is designed according to the requirements of the particular module and contains appropriate input fields, dropdown menus, buttons, date selectors, and validation options. A consistent form structure helps users enter information correctly and reduces confusion while working with different sections of the application.

The Login File is designed to provide controlled access to the system. It allows users to enter their username or email and password and authenticate themselves before accessing the application. Appropriate validation messages are provided when the login details are incorrect or incomplete. The form is designed to support the authentication and role-based access requirements of the system, allowing authorized users to access the functions relevant to their responsibilities.

The Product File is used to create and maintain product records in the inventory. It contains important fields such as SKU, product name, brand, category, subcategory, supplier, warehouse, cost price, selling price, current stock, reserved stock, reorder level, and maximum stock. Additional information such as monthly demand, lead time, product rating, and stock status can also be maintained. The form helps ensure that all required product information is captured in a structured manner.

The Purchase Order File is designed to create and manage purchase orders. It contains information such as PO number, supplier, warehouse, product, quantity, order date, expected delivery date, total value, payment status, delivery status, and priority. Users can update the order as it moves through different stages. The form provides a structured way of recording procurement activities and helps connect purchase information with inventory requirements.

The Stock Transfer File is used to record the movement of products between different warehouses. It includes transfer ID, product, SKU, source warehouse, destination warehouse, quantity, transfer date, expected arrival date, and transfer status. Users can select appropriate transfer statuses such as

The Returns File is used to record product return information and refund details. It contains fields such as return ID, invoice ID, product, customer, quantity, return reason, return date, refund amount, and return status. Users can select appropriate return reasons such as Damaged, Wrong Product, Defective, Customer Request, or Size Issue. This structured form ensures that returned products and associated financial information are properly recorded.

3.2 INPUT DESIGN

The Input Design of the Stock Level Prediction and Control Software focuses on collecting accurate and complete information from users. The input screens are designed to be simple, clear, and easy to understand so that users can enter information without confusion. Proper labels, text boxes, dropdown menus, date fields, numeric fields, and selection controls are used according to the type of data required. Validation is provided for important fields to reduce incorrect or incomplete entries.

The Login Input collects the username or email and password required for user authentication. The system validates the entered information before providing access to the application. This helps ensure that only authorized users can access protected modules and system information.

The Product Input is used to collect product-related information such as SKU, product name, brand, category, supplier, warehouse, cost price, selling price, current stock, reorder level, and maximum stock. Additional details such as monthly demand and lead time can also be entered. These inputs are stored and used by inventory and analytical modules.

The Supplier**,** Purchase, and Transfer Inputs collect information needed for procurement and warehouse operations. Supplier inputs include supplier details and performance-related information, while purchase order inputs include supplier, product, quantity, order date, delivery date, payment status, and priority. Stock transfer inputs record source warehouse, destination warehouse, product, quantity, transfer date, and transfer status.

The Sales and Returns Inputs are used to collect customer transaction and return information. Sales inputs include customer, product, quantity, price, discount, tax, payment method, and invoice details. Return inputs include invoice, product, customer, quantity, return reason, refund amount, and return status. These inputs allow sales and return information to be connected with inventory records.

Overall, the input design ensures that information entered into the system is accurate, consistent, complete, and easy to process. Validation and suitable input controls help reduce errors and ensure that reliable data is available for inventory calculations, reports, dashboard analytics, demand prediction, and automated reorder recommendations. The system validates the entered information before providing access to the application. This helps ensure that only authorized users can access protected modules and system information.

3.3 OUTPUT DESIGN

The Output Design of the Stock Level Prediction and Control Software focuses on presenting processed information to users in a clear, accurate, and understandable manner. The system converts stored inventory and transaction data into dashboards, tables, reports, charts, alerts, and summaries. The outputs are designed to help users monitor inventory conditions and make effective operational decisions.

The Dashboard Output provides a summarized view of important inventory and business information. It displays KPIs such as total products, stock units, inventory value, low-stock items, out-of-stock items, overstock items, pending purchase orders, sales, stock turnover rate, and forecast accuracy.

The Analytical Output presents information through charts and graphs. The system provides outputs for sales trends, inventory trends, category distribution, low-stock analysis, top-selling products, inventory value, supplier performance, stock movement, and AI demand forecasting. These visual outputs help users understand large amounts of data easily.

The Transaction Output provides detailed tables for products, suppliers, purchase orders, stock transfers, sales transactions, and returns. Information such as product details, quantities, dates, suppliers, warehouse locations, order status, payment status, and return status can be displayed in an organized format.

The AI Prediction Output displays current stock, historical average demand, predicted demand, reorder point, recommended order quantity, forecast accuracy, demand trend, and confidence score. This output helps users understand future stock requirements and supports better inventory planning.

The Report Output provides organized summaries of inventory, sales, purchases, suppliers, stock movements, and other operational activities. Reports can be used for monitoring, analysis, record keeping, and management decision making. The system also supports data export functionality for further use and analysis. This output helps users understand future stock requirements and supports better inventory planning.

Overall, the output design ensures that information generated by the system is clear, organized, accurate, and easy to interpret. The combination of dashboards, tables, charts, reports, alerts, and AI-based prediction outputs provides users with a complete view of inventory operations and supports effective stock management.

3.4 DATABASE DESIGN

The Database Design of the Stock Level Prediction and Control Software defines the structure used to store and manage all inventory-related information. The database acts as the central storage component of the system and maintains information about users, products, inventory, suppliers, warehouses, purchase orders, stock transfers, sales, and returns. A well-structured database helps the application retrieve information efficiently and maintain consistency between different modules.

The User Table stores authentication and user-related information required for system access. It can contain details such as user ID, name, email, password, role, and account status. The role information helps control access to different functions of the application.

The Product Table stores complete information about products. Important fields include Product ID, SKU, Product Name, Brand, Category, Subcategory, Supplier ID, Warehouse ID, Cost Price, Selling Price, Current Stock, Reserved Stock, Reorder Level, Maximum Stock, Monthly Demand, Lead Time, and Stock Status. This table acts as the main source of product information for inventory and analytical operations.

The Inventory Table maintains stock-related information for products. It records stock quantities and inventory movements such as stock received, stock sold, stock transferred, returns, and adjustments. This information is used to calculate available stock and identify different inventory health conditions.

The Supplier Table stores supplier information such as Supplier ID, Supplier Name, Contact, Location, Products Supplied, Total Orders, Pending Orders, On-Time Delivery Percentage, Quality Score, Total Purchase Value, and Supplier Status. This information supports supplier management and performance analysis.

The Warehouse Table stores information about warehouse locations and their inventory-related details. It can include Warehouse ID, Warehouse Name, Location, Total Products, Total Units, Inventory Value, Low Stock Count, Capacity Utilization, and Stock Movement. This structure supports the management of inventory across multiple locations.

The Sales Transaction Table stores POS and billing information such as Invoice ID, Date, Customer, Product, SKU, Quantity, Unit Price, Discount, Tax, Total Amount, Payment Method, Store, and Status. The Returns Table maintains return-related information such as Return ID, Invoice ID, Product, Customer, Quantity, Return Reason, Return Date, Refund Amount, and Status.

3.5 SYSTEM DEVELOPMENT

The System Development process involves converting the requirements and system design of the Stock Level Prediction and Control Software into a functional web-based application. The development focuses on integrating inventory management, product management, supplier management, warehouse operations, purchasing, sales, returns, analytics, and AI-based prediction into one centralized system. The application is developed in a modular manner so that each component can perform its specific function while sharing required information with other modules.

The development begins with the creation of the user interface, which provides login pages, dashboards, forms, tables, navigation menus, search options, and analytical charts. The interface is designed to be simple and user-friendly so that users can easily access inventory information and perform daily operations. Responsive layouts and consistent design elements are used to provide a professional appearance throughout the application.

The backend development handles the main application logic and connects the user interface with the database. It processes user requests, validates input, manages authentication and sessions, performs inventory calculations, updates stock information, and handles transactions. Different routes and functions are developed for products, inventory, suppliers, warehouses, purchase orders, transfers, sales, returns, reports, and other system operations.

The database development provides centralized storage for the application. Tables are created to maintain information about users, products, suppliers, warehouses, inventory, purchase orders, stock transfers, sales, and returns. Relationships between these records allow information to be shared between modules and help maintain data consistency across the application.

The inventory development focuses on updating stock whenever products are received, sold, transferred, returned, or adjusted. The system maintains current stock, reserved stock, available stock, reorder level, and maximum stock. Based on these values, products can be identified as healthy, low, critical, overstock, or out of stock.

The analytics and dashboard development converts stored inventory and transaction data into useful information. KPI cards, charts, tables, and inventory health indicators are generated from the underlying data. The dashboard presents information such as stock levels, inventory value, sales trends, top-selling products, supplier performance, and stock movement, helping users understand inventory performance more easily.

Additional development includes search, filtering, sorting, and pagination to support large datasets efficiently. Barcode and QR scanning, Excel import and export, reporting, authentication, and StockSense AI chatbot functionality are also integrated into the application. Finally, the developed system is tested to ensure that the modules work correctly and that information remains consistent.

3.5.1 DASHBOARD MODULE

The Dashboard Module provides a centralized view of the overall inventory status of the Stock Level Prediction and Control Software. It displays important stock information in a simple and user-friendly interface. The dashboard shows the total number of products available in the inventory along with current stock levels. It highlights low-stock and out-of-stock products to help users take immediate action. The module also presents predicted stock requirements based on previous sales and inventory data. Users can quickly view fast-moving and slow-moving products through graphical representations. Sales trends and stock movement can be monitored using charts and summary reports. A search and filter option allows users to find specific products easily. The dashboard provides real-time inventory updates whenever stock is added, sold, or modified. Automatic alerts and notifications are displayed when the stock reaches the minimum level. Authorized users such as managers and owners can access relevant inventory information according to their roles. The dashboard also helps reduce manual monitoring and saves time in inventory management.

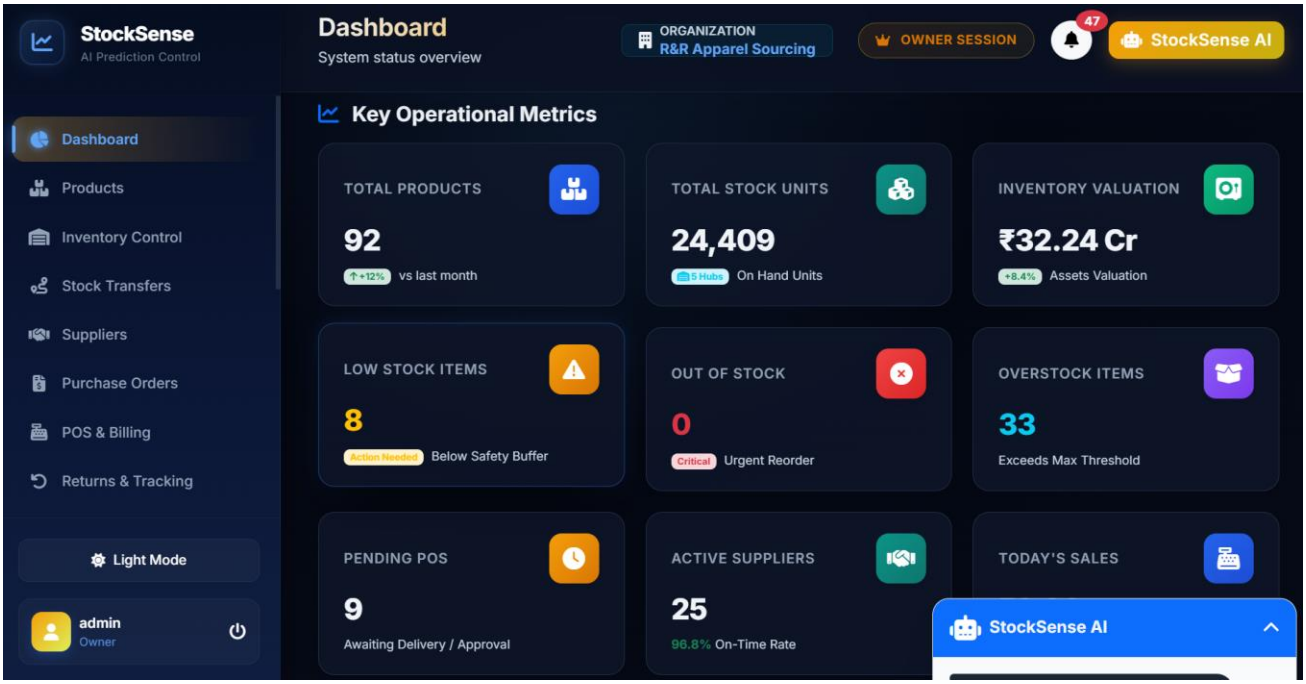


Fig 1. Dashboard Module

3.5.2 PRODUCTS MODULE

The Products Module is used to manage and maintain complete information about all products available in the inventory. It allows authorized users to add new products and enter details such as product name, product ID, category, quantity, price, supplier information, and minimum stock level. Users can view and update product details whenever required. The module automatically updates the available quantity whenever products are purchased or sold. It helps users identify low-stock and out-of-stock products quickly. A search and filter option is provided to find specific products based on their name, category, or product ID. The module can also support barcode or QR code scanning for faster product identification and inventory tracking. Product records are securely stored in the database for easy access and management. It reduces manual record keeping and minimizes errors in maintaining inventory information. The module works together with the prediction system to analyze product sales and estimate future stock requirements. Overall, the Products Module provides an organized and efficient way to manage product information and maintain accurate inventory levels. whenever products are purchased or sold. It helps users identify low-stock and out-of-stock products quickly. A search and filter option is provided to find specific products based on their name, category, or product ID. The module can also support barcode or QR code scanning for faster product identification and inventory tracking. Product records are securely stored in the database for easy access and management. It reduces manual record keeping and minimizes errors in maintaining inventory information.

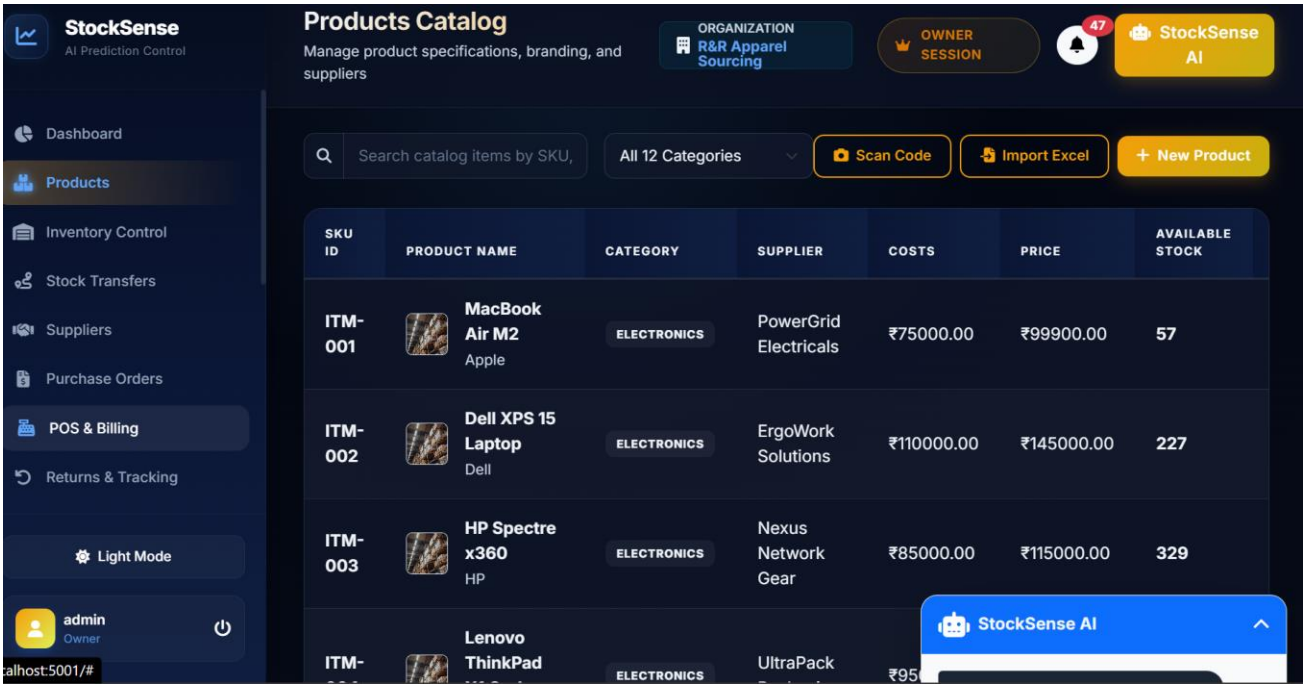


Fig 2. Dashboard Module

3.5.3 INVENTORY CONTROL MODULE

The Inventory Control Module is responsible for monitoring and controlling the stock levels of all products in the inventory. It keeps track of incoming and outgoing products and automatically updates the available stock quantity. The module helps users maintain the required stock level by comparing the current quantity with the predefined minimum and maximum stock levels. When the stock quantity reaches a low level, the system generates automatic alerts to notify the authorized users. It also identifies out-of-stock products and helps users take timely action to replenish them. The module records stock transactions such as purchases, sales, returns, and adjustments for accurate inventory tracking. Users can search and filter inventory records based on product name, category, quantity, or stock status. The system can use historical sales and inventory data to support stock-level prediction and better control. Barcode or QR code integration can make product identification and stock updating faster. All inventory activities are stored securely in the database for future reference and reporting. This module reduces manual stock checking, prevents overstocking and stock shortages, and improves the overall efficiency of inventory management.

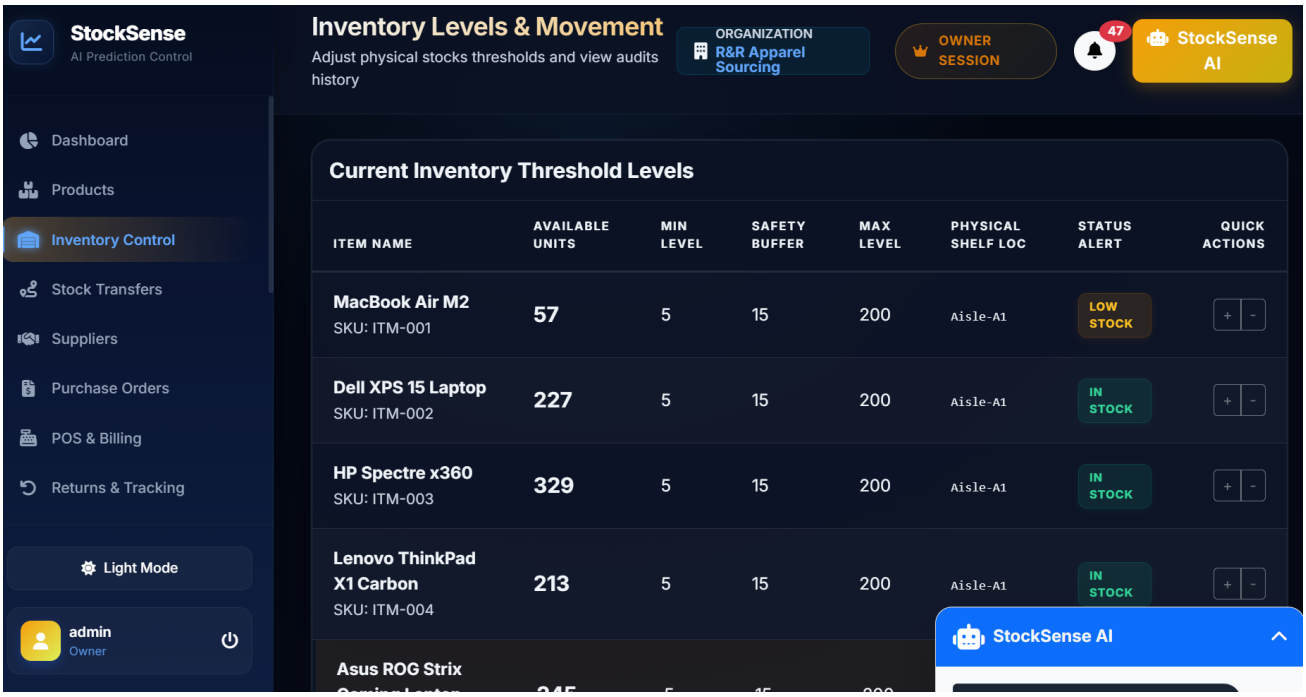


Fig 3. Inventory Control Module

3.5.4 STOCK TRANSFER MODULE

The Stock Transfers Module is used to manage the movement of products from one storage location, branch, or warehouse to another. It allows authorized users to create and manage stock transfer requests efficiently. The user can select the product, specify the quantity to be transferred, and choose the source and destination locations. The system checks the available stock at the source location before processing the transfer. Once a transfer is completed, the stock quantity is automatically reduced at the source location and added to the destination location. The module maintains a detailed record of every stock transfer, including product details, quantity, date, source, destination, and transfer status. Users can track whether a transfer is pending, approved, completed, or cancelled. Search and filter options help users quickly find specific transfer records. The system can also provide notifications for pending or completed transfers. Barcode or QR code scanning can be used to identify products and improve transfer accuracy. All transfer information is securely stored in the database for monitoring and future reference. This module helps prevent stock discrepancies, improves inventory visibility, and ensures smooth movement of products between different locations. product, specify the quantity to be transferred, and choose the source and destination locations. The system checks the available stock at the source location before processing the transfer. Once a transfer is completed, the stock quantity.

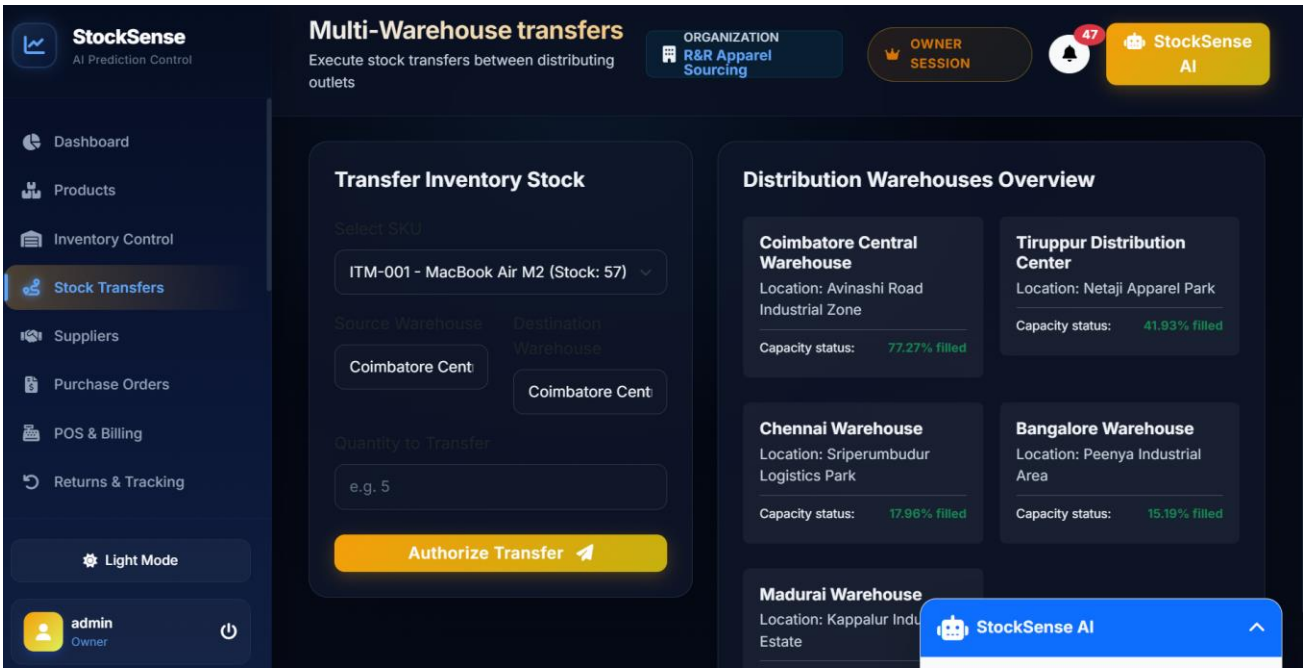


Fig 4. Stock Transfer Module

3.5.5 SUPPLIER MODULE

The Suppliers Module is used to manage and maintain complete information about all suppliers who provide products to the organization. It allows authorized users to add, view, update, and manage supplier details such as supplier name, supplier ID, contact number, email address, company name, and address. The module helps users maintain an organized record of suppliers and their associated products. Users can easily search and filter supplier information whenever required. It keeps track of products supplied by each supplier and supports better coordination during stock replenishment. The system can record purchase details, order quantities, delivery dates, and supplier transaction history. When stock levels become low, the supplier information can be used to support timely purchase and restocking decisions. The module helps monitor supplier performance based on delivery and order records. All supplier information is securely stored in the database for future reference. It reduces manual record keeping and improves the accuracy of supplier management. Overall, the Suppliers Module helps maintain reliable supplier relationships and ensures a smooth and efficient inventory replenishment process. details such as supplier name, supplier ID, contact number, email address, company name, and address. The module helps users maintain an organized record of suppliers and their associated products. Users can easily search and filter supplier information whenever it was required. The module helps monitor supplier performance based on delivery and order records in the supplier module.

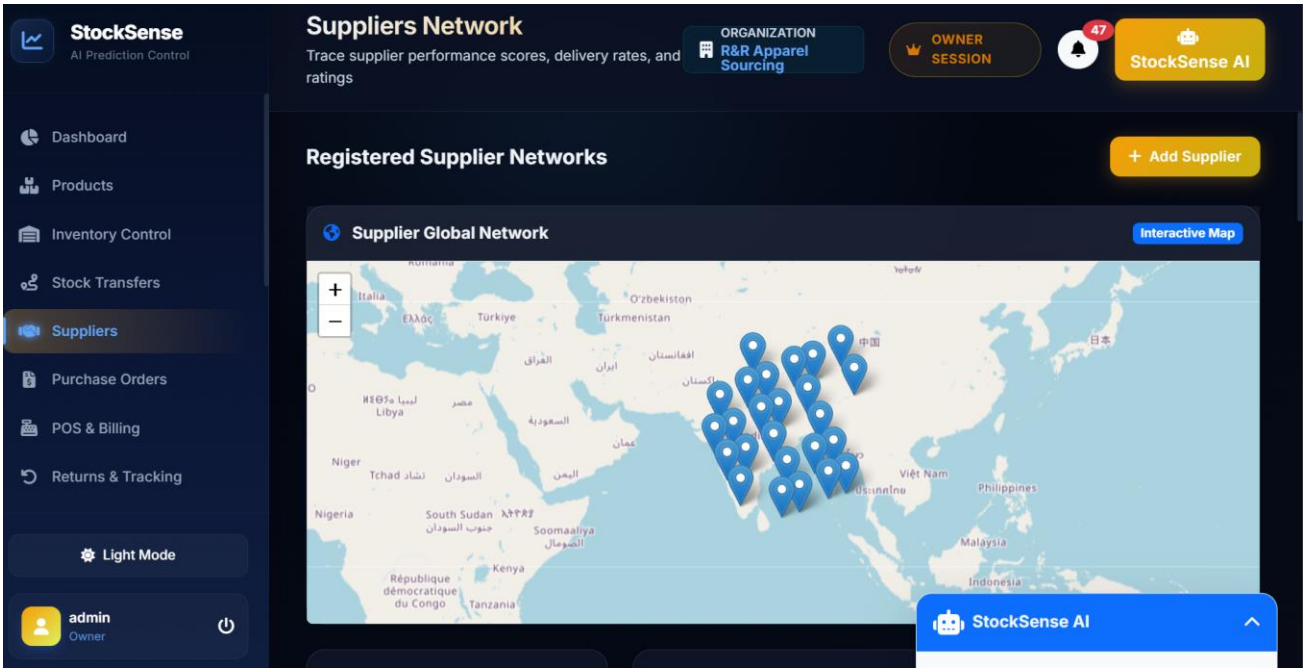


Fig 5. Supplier Module

3.5.6 PURCHASE ORDER MODULE

The Purchase Orders Module is used to create, manage, and track purchase orders for products required in the inventory. It allows authorized users to create a purchase order by selecting the supplier, products, required quantities, prices, and expected delivery date. The system helps users place orders when the available stock reaches the minimum stock level. Purchase orders can be assigned a unique order ID for easy identification and tracking. Users can view the status of each order, such as pending, approved, ordered, received, or cancelled. Once the ordered products are received, the system updates the inventory quantity automatically. The module maintains a complete history of purchase orders for future reference and reporting. Users can search and filter orders based on supplier, product, order date, or order status. It also helps monitor pending deliveries and avoid delays in stock replenishment. The purchase information is securely stored in the database and can be used for analyzing purchasing patterns. By reducing manual order processing, the module improves accuracy, saves time, and supports efficient inventory management. The Purchase Order Module is responsible for managing and tracking the purchase of products from suppliers. The module also helps identify pending, approved, processing, shipped, delivered, delayed, and cancelled purchase orders. By connecting purchase orders with inventory and supplier information, the module helps ensure that required products are purchased on time and that stock levels can be maintained efficiently.

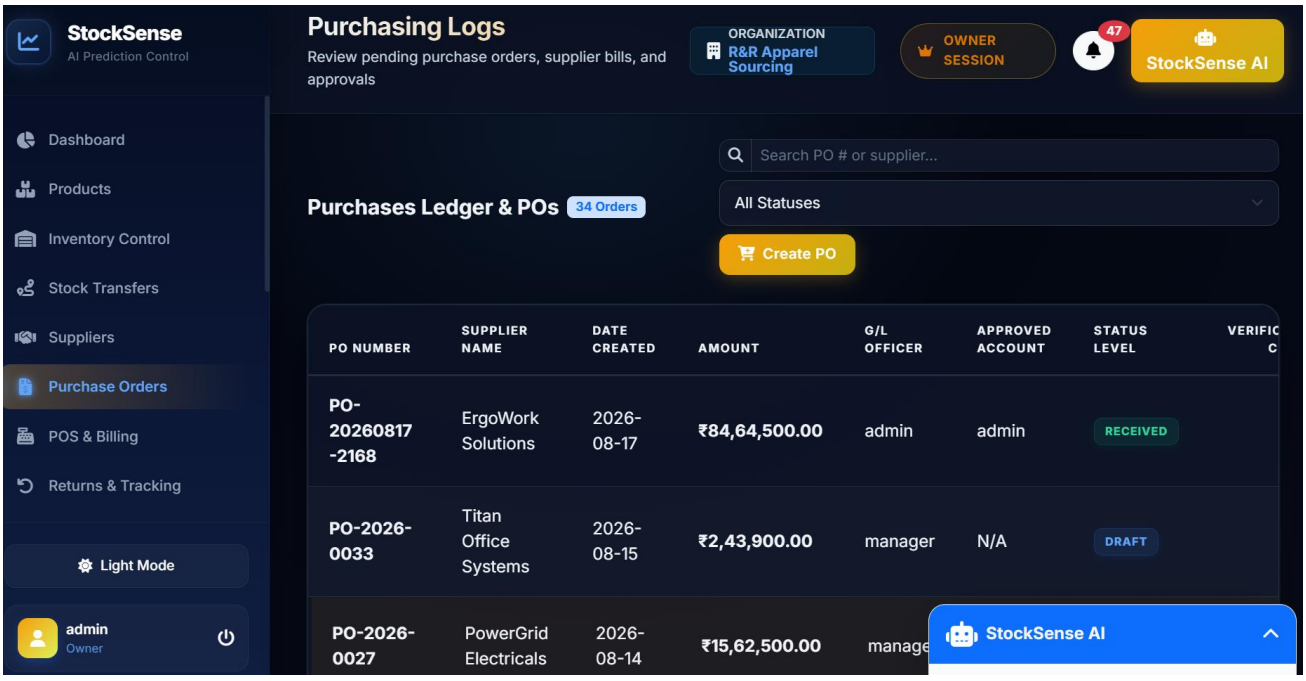


Fig 6. Purchase Order Module

3.5.7 POS & BILLING MODULE

The POS and Billing Module is used to manage sales transactions and generate bills quickly and accurately. It allows authorized users to select products, enter quantities, and calculate the total amount automatically. The system retrieves product prices and availability directly from the inventory database. Once a sale is completed, the corresponding stock quantity is automatically reduced from the inventory. The module calculates the subtotal, applicable discounts, taxes, and final payable amount. It generates a digital or printable bill containing details such as product name, quantity, price, total amount, date, and transaction ID. Users can search products easily using the product name, ID, barcode, or QR code. The module maintains a record of completed sales transactions for future reference and reporting. It also helps prevent billing errors by automatically performing calculations. Sales data generated through the POS system can be used by the prediction module to analyze customer demand and future stock requirements. The system securely stores transaction information in the database. Overall, the POS and Billing Module improves billing speed, maintains accurate sales records, and supports efficient inventory control. It allows users to record customer details, products, quantities, unit prices, discounts, taxes, total amounts, payment methods, and invoice information. The module processes sales transactions and generates billing details for customers. Once a sale is completed, the corresponding stock quantity can be updated in the inventory, helping maintain accurate stock records.

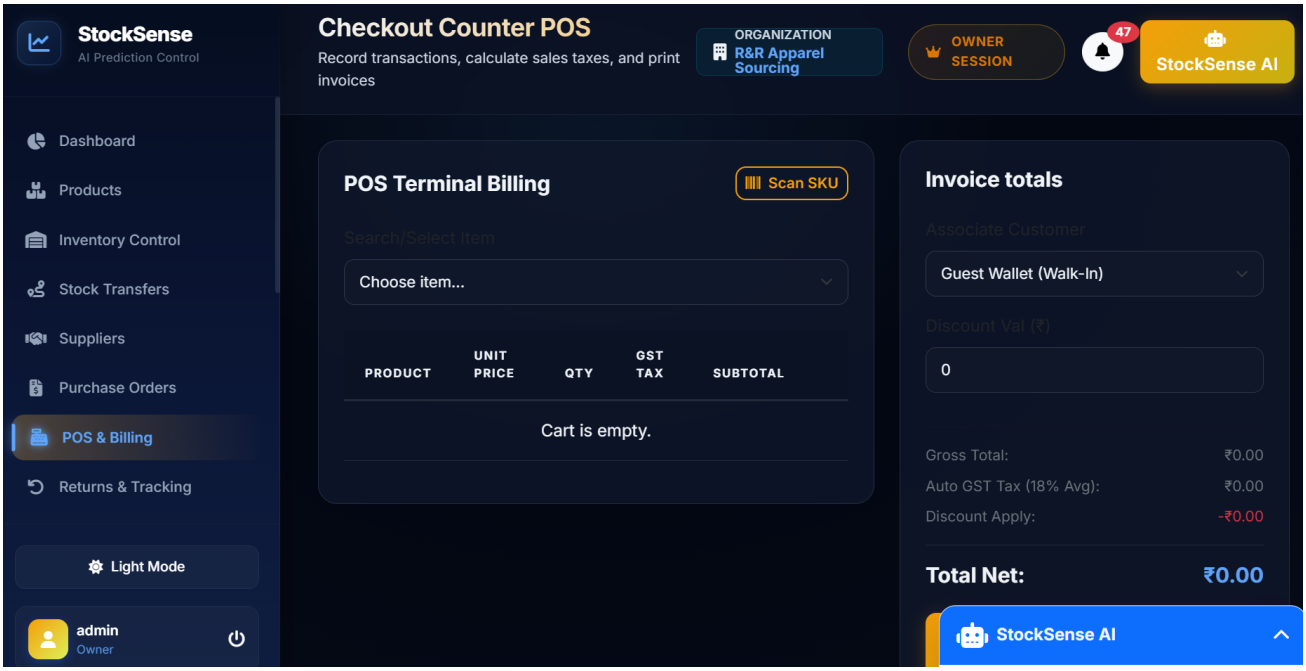


Fig 7. POS & Billing Module

3.5.8 RETURNS & TRACKING MODULE

The Returns & Tracking Module is used to manage returned products and track their movement throughout the inventory process. It allows authorized users to record product returns made by customers or sent back to suppliers. The module captures important details such as product ID, product name, quantity, return reason, date, transaction ID, and return status. When a valid customer return is accepted, the system can automatically update the available inventory quantity. Products that are damaged or defective can be marked separately so they are not added to usable stock. The module provides tracking information for returned products from the time the return is initiated until it is completed. Users can check whether a return is pending, approved, received, inspected, refunded, or rejected. Search and filter options help users quickly locate specific return records. The system maintains a complete return history for monitoring and reporting purposes. It also helps identify frequently returned products and possible inventory issues. All return and tracking information is securely stored in the database. Overall, this module improves return management, maintains accurate stock records, and provides better visibility of product movement. It stores details such as Return ID, Invoice ID, customer, product, quantity, return reason, return date, refund amount, and return status. The module helps users track returned items from the initial return request through verification and final processing. It can also update inventory information when a returned product is accepted and added back to available stock.

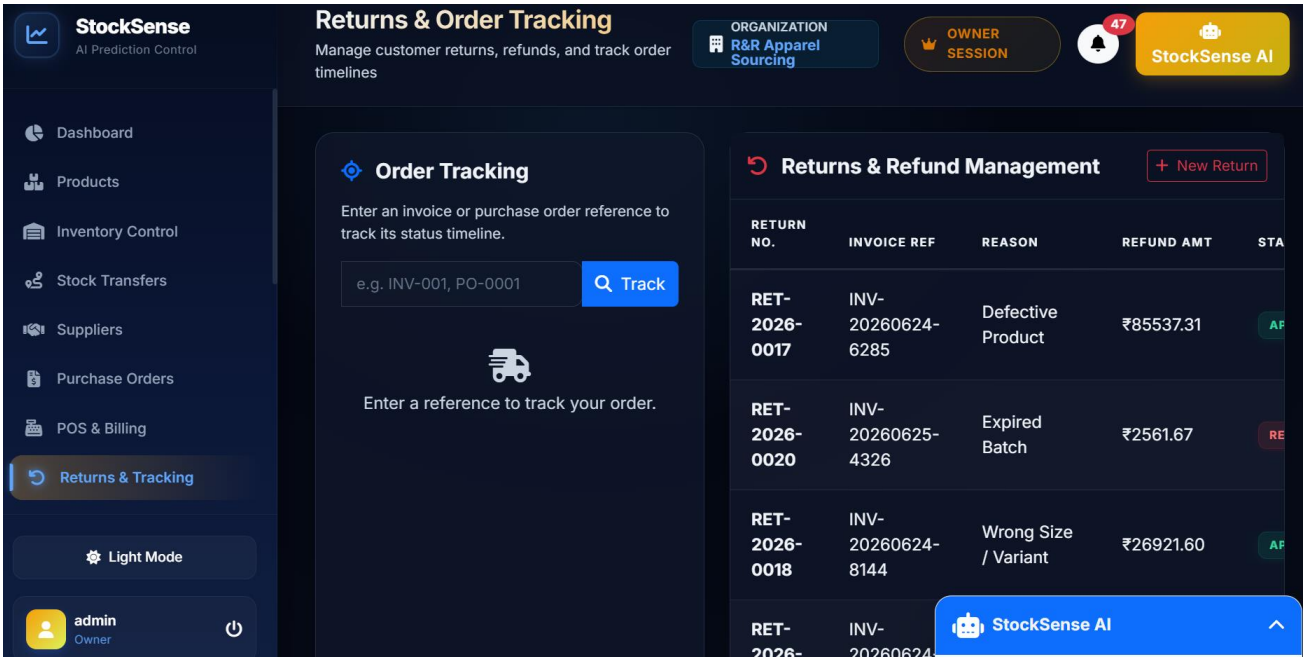


Fig 8. Return & Tracking Module

3.5.9 BARCODE SCANNING MODULE

The Barcode Scanning Module is used to identify and manage products quickly using barcode technology. It allows users to scan the barcode of a product using a barcode scanner or device camera. Once the barcode is scanned, the system automatically retrieves the corresponding product information from the database. Details such as product name, product ID, price, category, and available stock can be displayed instantly. The module helps users add products to sales transactions and generate bills more efficiently. It can also be used during stock receiving, inventory checking, stock transfers, and product returns. When a product is sold or received, the system automatically updates the stock quantity based on the scanned product. Barcode scanning reduces manual data entry and minimizes errors in product identification. It provides faster and more accurate inventory tracking, especially when handling a large number of products. The module maintains scanned transaction records for future reference and reporting. Overall, the Barcode Scanning Module improves operational efficiency, saves time, and provides accurate real-time product and inventory management. It helps users scan a product code and access details such as product name, SKU, category, price, supplier, and available stock. The module reduces manual product searching and data entry, making inventory operations faster and more efficient. It can be used during product receiving, sales, stock verification, warehouse operations, and stock transfers. By connecting barcode information with the product and inventory database, the module improves product identification, reduces entry errors, and supports efficient stock tracking.

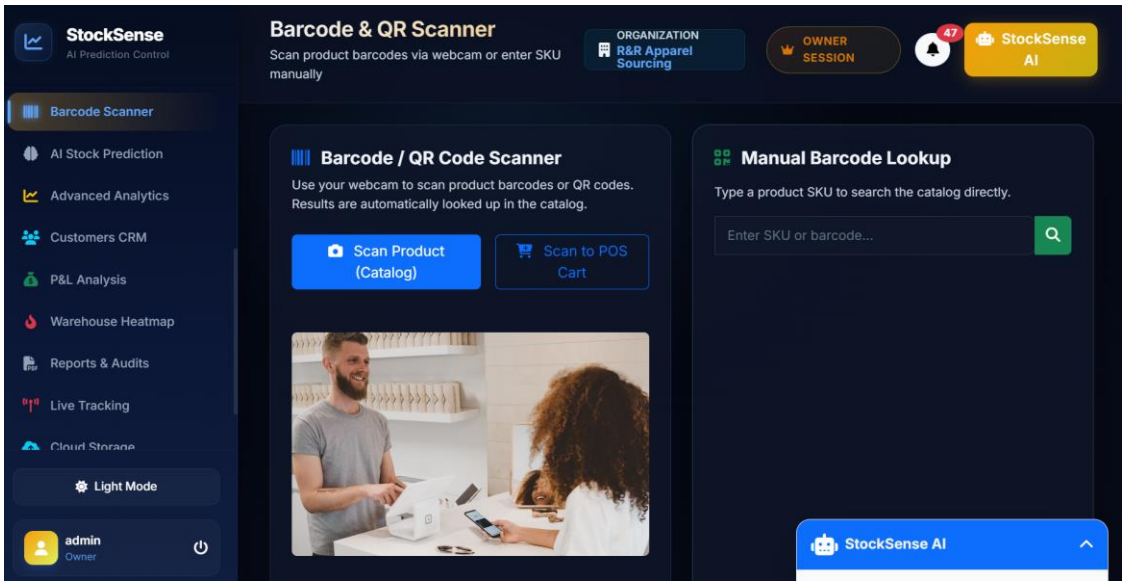


Fig 9. Barcode Scanner Module

3.5.10 AI STOCK PREDICTION MODULE

The AI Stock Prediction Module is used to predict future stock requirements by analyzing historical sales and inventory data. It uses artificial intelligence and machine learning techniques to identify patterns in product demand and estimate the quantity of stock required for future periods. The module considers factors such as previous sales, current stock levels, product demand, and sales trends to generate predictions. It helps users identify products that are likely to experience high or low demand.

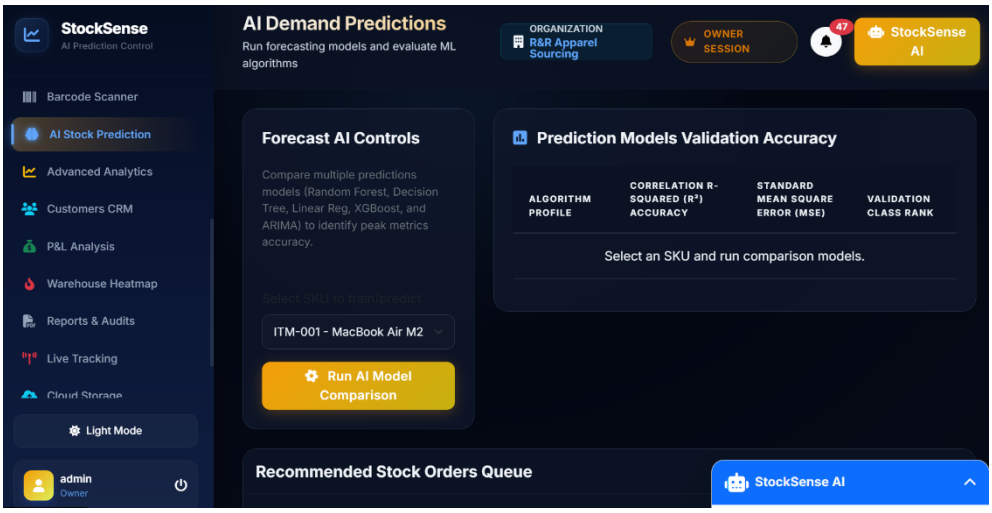


Fig 10. AI Stock Prediction Module

3.5.11 ADVANCED ANALYTICS MODULE

The Advanced Analytics Module is used to analyze inventory and sales data to provide meaningful insights for better decision-making. It collects and processes data such as sales history, stock levels, product demand, purchase records, and customer transactions. The module uses data analytics techniques to identify trends, patterns, and changes in inventory performance. It provides graphical reports and visualizations to make complex data easier to understand. Users can analyze fast-moving and slow-moving products and identify products with increasing or decreasing demand. The module can compare sales performance across different time periods, products, and categories. It also helps identify overstocked and understocked products by analyzing current inventory conditions. Advanced analytics can support demand forecasting and assist the AI prediction system in generating more accurate stock recommendations. Users can apply search, filtering, and date-based analysis to obtain specific information from the database. The module generates useful reports that help managers understand inventory performance and make informed purchasing decisions. By

converting raw inventory data into actionable insights, the Advanced Analytics Module improves planning, reduces inventory-related costs, and supports efficient overall inventory management.

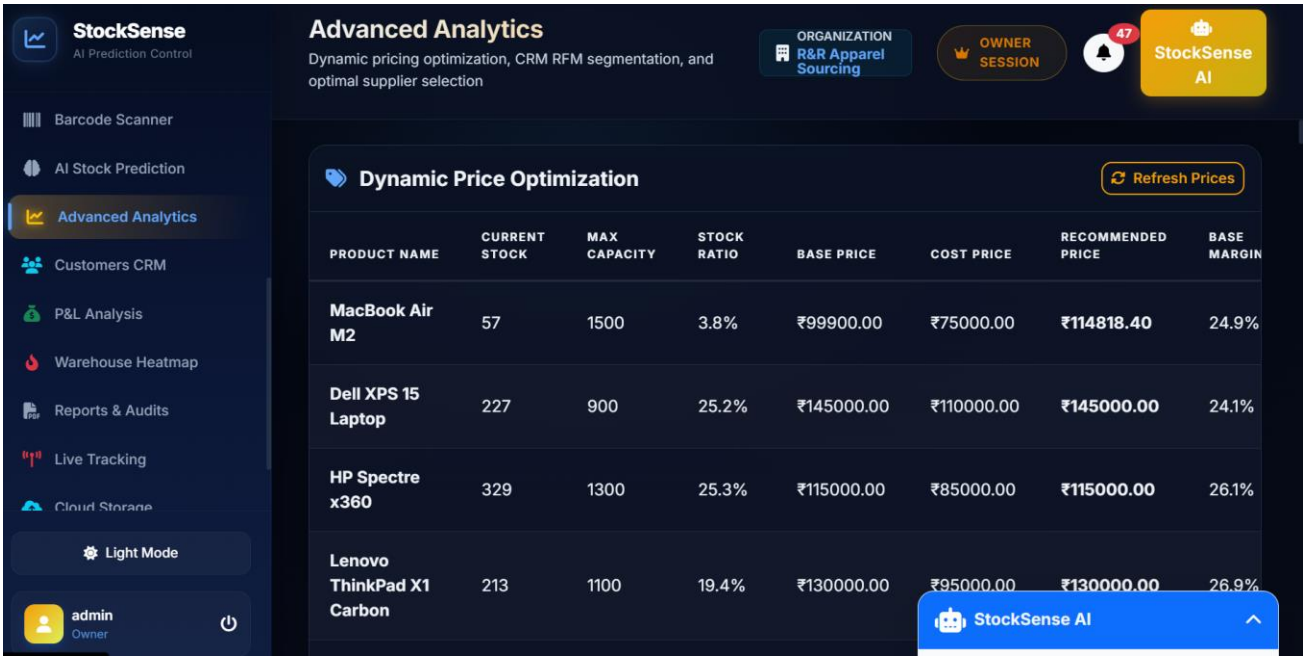


Fig 11. Advanced Analytics Module

3.5.12 CUSTOMERS CRM MODULE

The Customer CRM Module is used to manage and maintain customer information and their interactions with the business. It allows authorized users to add, view, update, and manage customer details such as customer ID, name, contact information, purchase history, and transaction details. The module maintains a centralized customer database that helps users access customer information easily. It records customer purchase activities and helps identify frequently purchased products and customer preferences. Users can search and filter customer records based on customer name, ID, contact details, or purchase history. The module can provide insights into customer buying patterns by analyzing previous sales transactions. It helps businesses improve customer service by providing quick access to relevant customer information. Customer purchase history can also support personalized offers, discounts, and promotional activities. The system securely stores customer records and maintains accurate transaction information for future reference. By connecting customer information with sales and inventory data, the module provides useful insights for business planning. Overall, the Customer CRM Module improves customer relationship management, supports better service, and helps increase customer satisfaction and retention. It helps businesses improve customer service by providing quick access to relevant customer information.

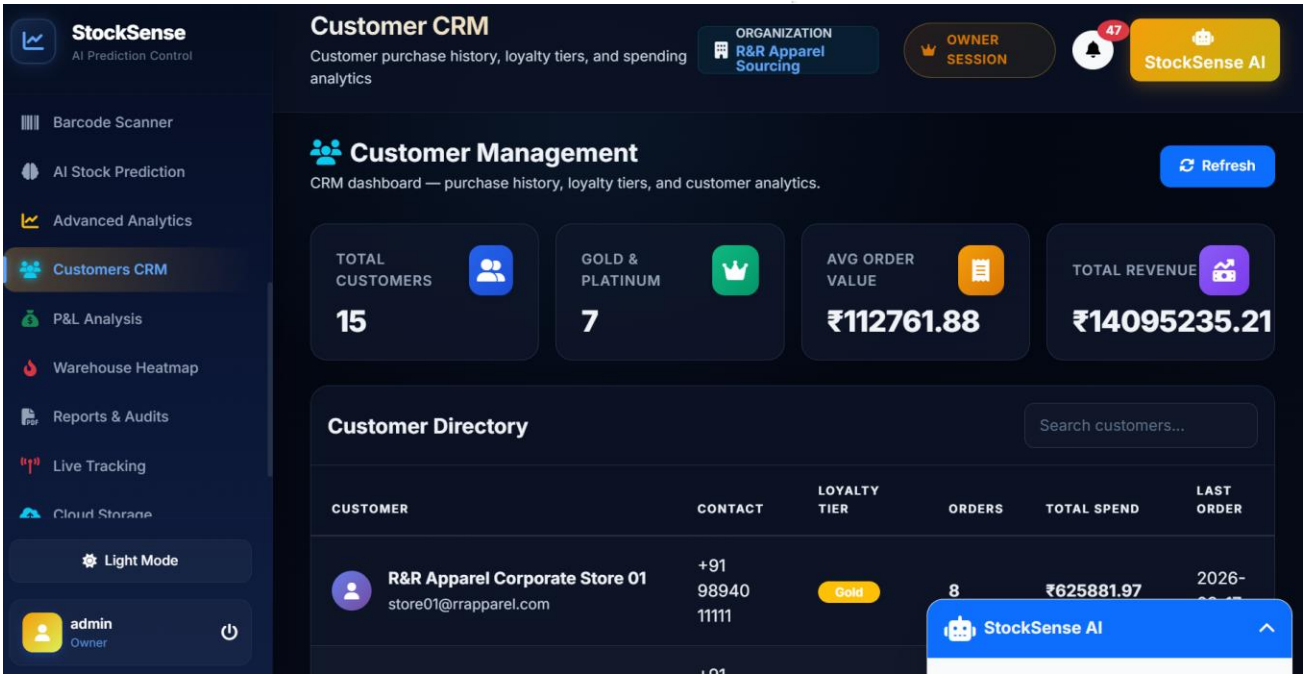


Fig 12. Customer CRM Module

3.5.13 USER AUTHENTICATION MODULE

The User Authentication Module is used to provide secure access to the Stock Level Prediction and Control Software. It allows users to log in to the system using their registered username or email and password. The module verifies the entered login details before providing access to the application. It supports role-based access for different users such as administrators, managers, and authorized staff. Each user is provided with permissions based on their assigned role, ensuring that sensitive inventory information is protected. The module prevents unauthorized users from accessing or modifying important product, sales, supplier, and stock records. Passwords and user credentials are securely managed to improve system security. The system can also provide logout functionality to safely end an active session. Invalid login attempts are rejected and appropriate error messages are displayed to the user. User account information is securely stored in the database. By providing controlled and secure access, the module protects business data and reduces the risk of unauthorized activities. Overall, the User Authentication Module ensures secure, reliable, and role-based access to the inventory management system. The module prevents unauthorized users from accessing or modifying important product, sales, supplier, and stock records. Passwords and user credentials are securely managed to improve system security

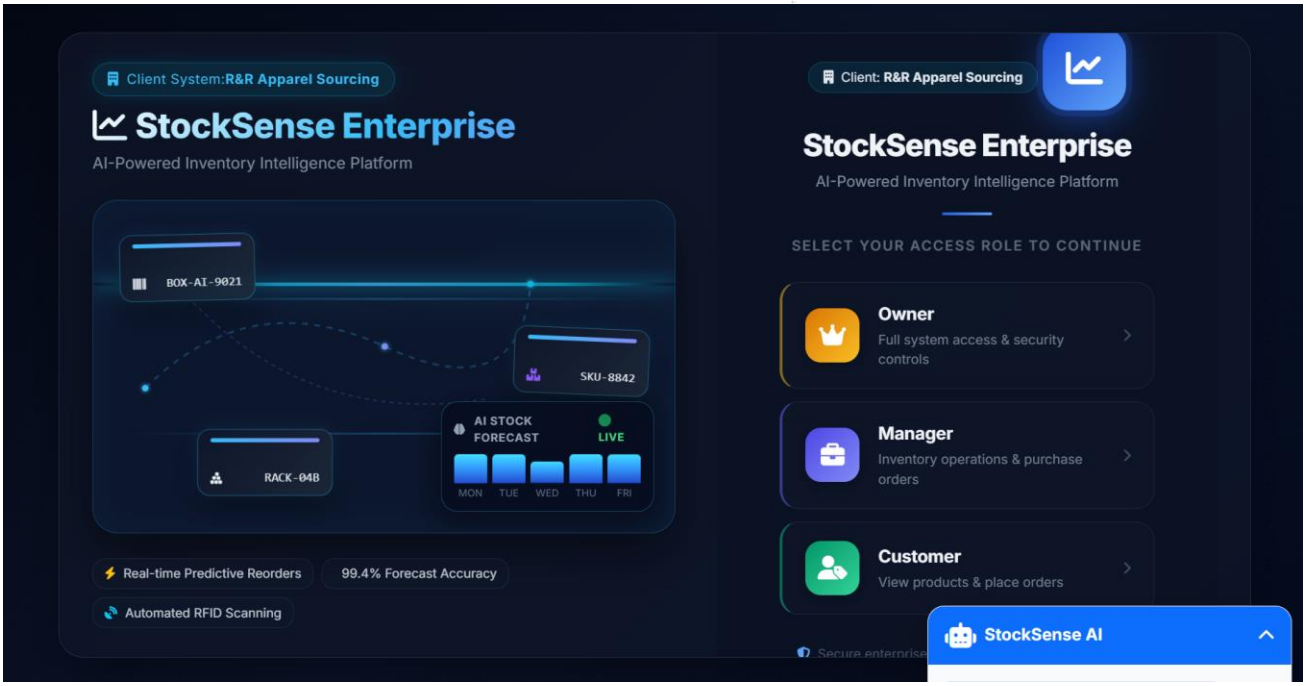


Fig 13. User Authentication Module

3.5.14 P&L ANALYSIS MODULE

The P&L (Profit and Loss) Analysis Module is used to analyze the financial performance of the business by comparing total sales revenue with the costs and expenses associated with inventory management. It collects information such as sales transactions, purchase costs, product prices, discounts, and other relevant expenses from the system database. The module calculates the overall revenue, total cost, gross profit, net profit, and loss for a selected period. Users can analyze financial performance on a daily, weekly, monthly, or yearly basis. It provides charts, graphs, and summary reports to present financial information in an easy-to-understand format. The module helps identify products that generate higher profits and products that contribute to lower profitability. Users can also compare profit and loss across different products, categories, and time periods. The analysis can support better purchasing and pricing decisions by providing clear financial insights. It helps managers monitor business performance and identify areas where costs can be reduced. All financial data and analysis results are securely maintained in the database. Overall, the P&L Analysis Module provides useful financial insights and supports effective business and inventory management decisions. It provides charts, graphs, and summary reports to present financial information in an easy-to-understand format. The module helps identify products that generate higher profits and products that contribute to lower profitability.

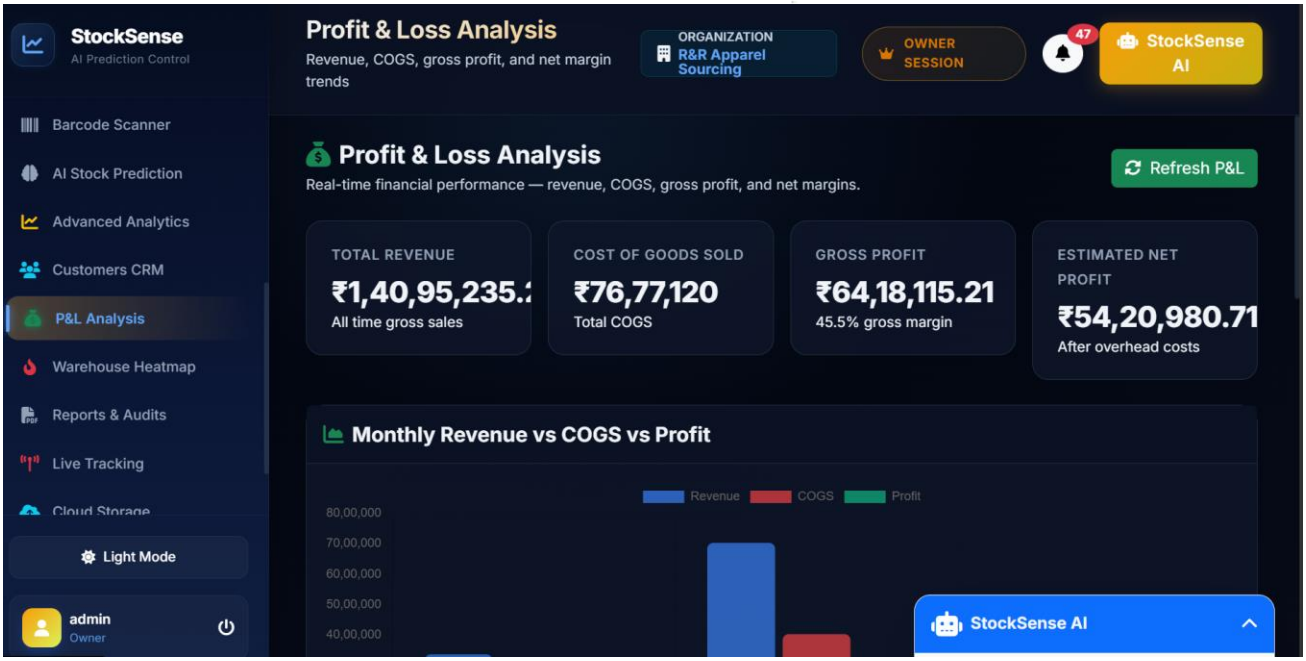


Fig 14. P&L Analysis Module

3.5.15 LIVE TRACKING MODULE

The Live Tracking Module is used to monitor the real-time movement and status of products within the inventory system. It allows authorized users to track products from the time they are received until they are sold, transferred, or returned. The module provides updated information about product quantity, location, movement status, and transaction activity. It can track stock movements between warehouses, branches, and storage locations. Barcode or QR code scanning can be integrated to identify products and update their movement information quickly. The system automatically records important details such as product ID, quantity, source location, destination location, date, and time. Users can view current stock movements through the dashboard and monitor ongoing transfers. The module also helps identify delays, missing products, or unexpected changes in stock levels. Search and filter options allow users to track specific products or transactions easily. All tracking information is stored securely in the database for future reference and reporting. By providing real-time visibility of inventory movement, the module improves stock accuracy and operational control. Overall, the Live Tracking Module helps businesses monitor products efficiently and maintain better control over their inventory. It can track stock movements between warehouses, branches, and storage locations. Barcode or QR code scanning can be integrated to identify products and update their movement information quickly.

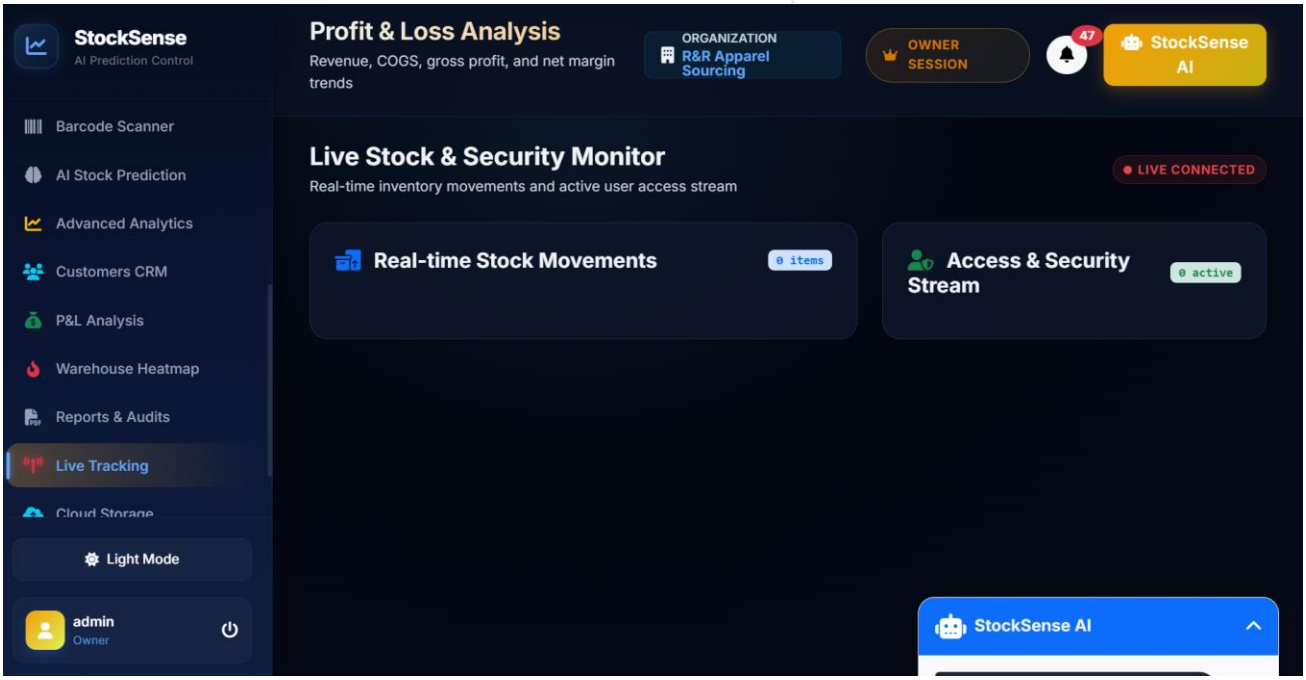


Fig 15. Live Tracking Module

3.5.16. MANAGER DASHBOARD MODULE

The Manager Dashboard provides managers with a centralized view of important inventory, sales, and operational information. It displays key details such as total products, available stock, low-stock items, out-of-stock products, sales, purchase orders, and stock movements. The dashboard helps managers monitor the overall performance of inventory activities in real time. It provides visual charts and graphs to show sales trends, stock levels, product demand, and purchasing activities. Managers can quickly identify products that require restocking and review AI-based stock predictions for future demand. The dashboard also displays alerts and notifications for low stock, pending purchase orders, stock transfers, and other important activities. Search and filter options allow managers to view specific products, categories, or time periods. It provides access to reports such as sales analysis, inventory performance, and profit and loss analysis. The dashboard also helps managers monitor staff activities and ensure that inventory operations are performed correctly. All information is retrieved from the centralized database and updated according to system transactions. Overall, the Manager Dashboard supports efficient monitoring, faster decision-making, and effective control of inventory and business operations. It displays key details such as total products, available stock, low-stock items, out-of-stock products, sales, purchase orders, and stock movements. The dashboard helps managers monitor the overall performance of inventory activities in real time.



Fig 16. Manager Dashboard Module

3.5.17. CUSTOMER MANAGEMENT

The Customer Management Module is used to manage and maintain complete information about customers in the inventory and sales management system. It allows authorized users to add, view, update, and delete customer records whenever required. The module stores important details such as customer ID, name, contact number, email address, address, and purchase history. It provides a centralized database for maintaining accurate customer information and makes customer records easy to access. Users can search and filter customers based on their name, ID, contact details, or transaction history. The module records customer purchases and helps track their buying patterns and frequently purchased products. It supports better customer service by providing quick access to previous transaction details. Customer information can also be used to provide personalized offers, discounts, and promotional activities. The system securely stores customer records and protects them from unauthorized access. It can work with the POS and Billing Module to automatically associate sales transactions with the appropriate customer. It allows authorized users to add, view, update, and delete customer records whenever required. The module stores important details such as customer ID, name, contact number, email address, address, and purchase history. It provides a centralized database for maintaining accurate customer information and makes customer records easy to access. The system securely stores customer records and protects them from unauthorized access.

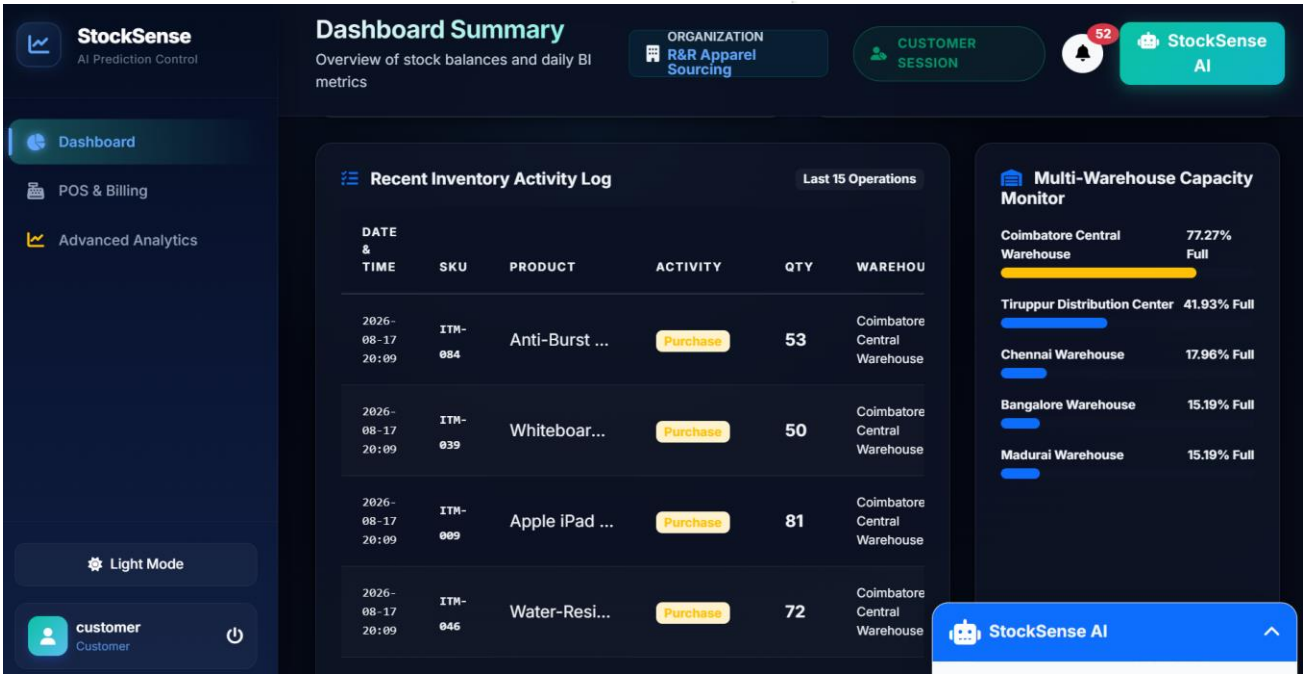


Fig 17. Customer Management Module

3.5.18. LOGOUT

The Logout Module is used to securely end the user's active session in the Stock Level Prediction and Control Software. It allows users to exit the system safely after completing their activities. When the logout option is selected, the system terminates the current user session and prevents unauthorized access to the account. The user is redirected to the login page after logging out successfully. This module is especially important when multiple users access the system from the same device. It helps protect sensitive information such as product details, inventory records, customer data, sales transactions, and financial reports. The system ensures that the user's session credentials are no longer active after logout. Users must provide valid login credentials again to access the system. The logout function is available to authorized users according to their role and access permissions. Overall, the Logout Module improves system security, protects user data, and ensures that user sessions are properly terminated. This prevents unauthorized users from accessing the application through an active session on the device. The module provides a simple and accessible logout option within the application interface and helps maintain controlled access to protected features such as inventory, purchase orders, sales, reports, and analytics. By properly ending the user session, the module improves the security and reliability of the overall system. When the user selects the logout option, the active session is terminated and the user is redirected to the login page.

4. SOFTWARE TESTING AND IMPLEMENTATION

4.1 TESTING

Testing is an important stage in the development of the Stock Level Prediction and Control Software. It is carried out to verify that the different modules of the application work according to the specified requirements and produce the expected results. Testing helps identify errors in functionality, input validation, database operations, navigation, calculations, and user interaction before the system is used for demonstration or practical operation.

The testing process is performed on the major modules of the application, including User Authentication, Product Management, Inventory Control, AI Demand Prediction, Automated Reorder, Supplier and Purchase Management, Warehouse and Stock Transfer, and Sales, Billing and Returns. Additional testing is performed on dashboard displays, search and filtering, reports, barcode/QR functionality, and database operations. Both valid and invalid inputs are considered to ensure that the application responds correctly under different conditions.

The objective of testing is to verify that data entered through the forms is correctly processed and stored in the database, that inventory quantities are updated properly after transactions, and that the dashboard and reports display consistent information. Testing also checks whether the AI prediction and reorder features produce appropriate results based on the available inventory and demand data. The overall testing process helps improve the reliability, accuracy, usability, and consistency of the application.

4.1.1. TESTING METHODS

(i) Unit Testing

Unit testing is performed on individual functions and components of the application. Functions related to login validation, product entry, stock calculation, database operations, and prediction processing are tested independently to verify that each component works correctly.

(ii) Integration Testing

Integration testing checks whether different modules work correctly when connected with one another. For example, when a sales transaction is completed, the corresponding inventory quantity should be updated in the database and reflected in the dashboard.

4.2 TEST CASES

Test Case ID	Module	Expected Result	Status
TC01	Login	Valid user should login successfully	Pass
TC02	Products	Product should be added correctly	Pass
TC03	Inventory	Stock quantity should update correctly	Pass
TC04	Suppliers	Supplier details should be stored	Pass
TC05	Purchase Order	Purchase order should be created	Pass
TC06	Stock Transfer	Stock should transfer between warehouses	Pass
TC07	POS & Billing	Sales transaction should be recorded	Pass
TC08	Returns	Return details should be recorded	Pass
TC09	AI Prediction	Demand prediction should be displayed	Pass
TC10	Reorder	Reorder recommendation should be generated	Pass
TC11	Dashboard	Updated information should be displayed	Pass
TC12	Logout	User should be logged out successfully	Pass

5. CONCLUSION

The Stock Level Prediction and Control Software has been developed as an integrated inventory management and intelligence platform for managing the complete flow of inventory-related activities. The system brings product management, inventory control, suppliers, warehouses, purchasing, sales, stock transfers, returns, reporting, and analytics together within a centralized application. This integrated approach reduces the dependence on separate records and provides users with a more organized and convenient method of managing inventory information.

The system provides a professional Dashboard Module that gives users a summarized view of important inventory and business activities. KPI cards, inventory health indicators, sales trends, stock balance, and analytical charts help users understand the current condition of the organization. The dashboard also provides access to functions such as AI Demand Prediction, Barcode/QR Scanning, Advanced Analytics, Reports and Audits, Live Tracking, Cloud Storage, and other operational features. The inclusion of these features makes the system more informative and suitable for continuous monitoring.

The Product Management and Inventory Control functions provide a structured method of maintaining product details and monitoring stock quantities. The system maintains information such as SKU, product name, category, supplier, warehouse, pricing, stock quantity, reorder levels, and demand-related information. Inventory can be classified according to different conditions, including healthy, low, critical, overstock, and out-of-stock levels. This allows users to identify inventory problems at an early stage and take suitable corrective action.

The project also integrates supplier management, purchase orders, warehouse management, stock transfers, POS and billing, and returns tracking. These modules create a connected flow of inventory information from purchasing and warehouse storage to sales and product returns. Multiple warehouse support allows inventory to be monitored across different locations, while purchase and supplier information helps users manage procurement activities more systematically.

One of the major achievements of the system is the integration of AI Demand Prediction and Automated Reorder Recommendations. Historical demand information can be used to estimate future product requirements, while the system provides information such as predicted demand, reorder points, recommended order quantities, forecast accuracy, demand trends, and confidence scores. Automated reorder recommendations further support replenishment decisions by identifying products that require

attention. These functions provide an additional level of analytical support beyond conventional inventory monitoring.

The system also provides search, filtering, sorting, and pagination facilities, making it practical to work with larger datasets. Users can search products and records using relevant fields and filter information according to warehouse, supplier, stock status, and date. These facilities improve usability and reduce the time required to locate specific information.

Another important aspect of the project is data consistency. Inventory-related values and dashboard information are designed to follow defined relationships and calculations. Available stock, inventory value, profit, low-stock status, out-of-stock status, and overstock conditions are based on underlying inventory data. Dashboard KPIs, charts, tables, and reports are intended to use the same dataset so that the information presented across different modules remains consistent and meaningful.

The project also demonstrates the practical use of Computer Science, Database Management, Web Development, Data Analytics, Data Visualization, and Artificial Intelligence in a single application. Features such as authentication, barcode/QR scanning, Excel import and export, reports, StockSense AI chatbot, and other existing modules extend the functionality beyond basic stock recording.

In conclusion, the Stock Level Prediction and Control Software provides a comprehensive environment for inventory monitoring, analysis, and control. It combines operational management with analytical and intelligent features to provide better visibility into stock conditions and support informed inventory decisions. The complete system provides a strong foundation for an enterprise-level inventory intelligence platform and demonstrates how modern software and data-driven techniques can be applied to improve inventory management.

5.1 FUTURE ENHANCEMENT

The Stock Level Prediction and Control Software can be further enhanced by introducing more advanced artificial intelligence and machine learning capabilities. The current demand prediction functionality can be extended to analyze larger historical datasets and identify seasonal patterns, changes in customer demand, product popularity, and sales fluctuations. More advanced forecasting models can be introduced to improve prediction accuracy for individual products and categories. The system can also compare predicted demand with actual sales and continuously improve future predictions based on the difference between forecasted and actual values.

The system can also be enhanced with a more advanced automated inventory replenishment mechanism. Instead of only displaying reorder recommendations, future versions can automatically prepare purchase recommendations based on predicted demand, current stock, reorder level, supplier lead time, and warehouse availability. The system can consider the quantity already available in other warehouses before recommending a new purchase. This can help reduce unnecessary purchases and improve the distribution of stock across different locations.

A real-time inventory monitoring system can be introduced to provide continuous updates whenever an inventory transaction occurs. Stock received, stock sold, stock returned, warehouse transfers, and inventory adjustments can immediately update the dashboard and related analytical views. Managers can receive instant notifications whenever critical stock conditions, unusual transactions, or delayed purchase orders are detected. This can make the application more responsive and useful for day-to-day operational monitoring.

The application can be extended into a cloud-based inventory management platform so that authorized users can access the system from different locations. Cloud deployment can provide centralized data storage, easier system updates, improved scalability, and remote access. A mobile version can also be developed for managers, warehouse employees, and sales staff. Mobile users could check stock availability, scan products, receive notifications, view reports, and update inventory information directly from mobile devices.

Future versions can provide an improved notification and alert system with multiple notification channels. Email, SMS, mobile notifications, and in-application alerts can be used for low-stock warnings, critical-stock conditions, purchase order delays, pending approvals, failed transactions, and other important events. Users could customize which notifications they receive according to their role and responsibilities. This would make the system more personalized and reduce the possibility of missing important inventory activities.

The Barcode and QR Code module can also be expanded to support complete warehouse operations. Future development can include barcode-based goods receiving, stock verification, shelf identification, warehouse transfer confirmation, product picking, and dispatch tracking. QR codes can be used to provide instant access to product information, stock status, supplier details, and transaction history. This can reduce manual entry and improve the speed of warehouse activities.

The system can further include advanced business intelligence and reporting capabilities. Users could create customized reports by selecting specific dates, products, warehouses, suppliers, categories, or

transaction types. Reports could include inventory turnover, slow-moving products, fast-moving products, profit and loss, supplier performance, warehouse efficiency, sales growth, purchase trends, and customer activity. Interactive reports with drill-down functionality could allow users to move from a summary value to the detailed records supporting that value.

Another future enhancement is API integration with external business systems. The application can be connected with accounting software, ERP platforms, e-commerce websites, billing applications, payment systems, and logistics services. This would allow data such as products, orders, customers, payments, and inventory quantities to move automatically between systems. Such integration can reduce duplicate data entry and provide a more connected business environment.

The security framework can also be strengthened in future versions. Additional security measures such as two-factor authentication, stronger password policies, session monitoring, detailed audit logs, permission management, login activity tracking, and account lockout mechanisms can be incorporated. Sensitive operations such as changing stock quantities, deleting records, modifying user permissions, or approving purchase orders can require additional authorization.

Future versions can introduce AI-based anomaly detection for inventory and sales data. The system can identify unusual stock reductions, unexpected sales increases, repeated returns, suspicious transaction patterns, and differences between expected and actual inventory. Such alerts can help users investigate potential errors or unusual activities at an early stage.

The Customer CRM can also be enhanced with intelligent customer analytics. The system can analyze purchase history, frequently purchased products, purchase frequency, average order value, and customer activity. These insights can support customer segmentation and help organizations understand buying patterns. The information can also contribute to demand forecasting by identifying changes in customer purchasing behavior.

The Warehouse Heatmap can be expanded into an intelligent warehouse optimization feature. It can show product movement, stock concentration, frequently accessed products, available storage capacity, and slow-moving inventory. In the future, the system could recommend suitable warehouse locations for products based on demand frequency and movement patterns, helping improve warehouse efficiency.

The application can also introduce automatic backup and disaster recovery features. Regular backups of the database and important reports can be created automatically and stored securely. In case of

system failure or data corruption, authorized administrators can restore the latest available backup. This will improve data reliability and reduce the risk of losing important inventory and transaction information.

A voice-enabled StockSense AI assistant can be introduced as another advanced feature. Users could ask questions such as “How many units are available?”, “Which products are low in stock?”, “What are today's sales?”, or “Which purchase orders are delayed?” using natural language. The assistant could provide responses based on the available system data, making the application easier to use for users who prefer conversational interaction.

The system can also support personalized dashboards in the future. Managers, warehouse staff, sales users, and other roles could have dashboards containing only the KPIs and information relevant to their responsibilities. Users could select preferred charts, widgets, filters, and report views. This would make the interface more flexible and improve the overall user experience.

Finally, the system can be extended into a complete enterprise inventory intelligence platform by continuously improving its AI models, analytics, integrations, security, and automation capabilities. The existing foundation already combines inventory management, analytics, prediction, reporting, and operational modules, while future enhancements can focus on cloud deployment, mobile access, intelligent automation, advanced security, real-time monitoring, and integration with external business applications. This would allow the system to support larger datasets, more users, multiple locations, and more complex inventory operations while maintaining a professional and scalable architecture.

The system can be further developed with real-time analytics and cloud deployment so that authorized users can access inventory information from different locations and devices. A mobile application can also be introduced for stock checking, barcode scanning, warehouse activities, and notifications. Email, SMS, and application alerts can be added for low-stock conditions, critical inventory, delayed purchase orders, and important system activities.

Future enhancements can also include advanced barcode and QR-code tracking, improved warehouse management, business intelligence reports, API integration with ERP and e-commerce systems, and stronger security features such as two-factor authentication and detailed audit logs. AI-based anomaly detection can be used to identify unusual stock movements, sales patterns, and inventory mismatches. enhancements can transform the application into a more advanced AI-driven and cloud-based inventory intelligence platform with improved automation, real-time monitoring, remote accessibility.

BIBLIOGRAPHY

[1]. PYTHON DOCUMENTATION – PYTHON PROGRAMMING LANGUAGE.
https://docs.python.org/3/

[2]. FLASK DOCUMENTATION – FLASK WEB FRAMEWORK.
https://flask.palletsprojects.com/

[3]. SQLITE DOCUMENTATION – SQLITE DATABASE.
https://www.sqlite.org/

[4]. SQLALCHEMY DOCUMENTATION – PYTHON SQL TOOLKIT AND ORM.
https://www.sqlalchemy.org/

[5]. BOOTSTRAP DOCUMENTATION – RESPONSIVE WEB DEVELOPMENT FRAMEWORK.
https://getbootstrap.com/

[6]. MDN WEB DOCS – HTML, CSS, JAVASCRIPT AND WEB TECHNOLOGIES.
https://developer.mozilla.org/

[7]. PANDAS DOCUMENTATION – DATA ANALYSIS AND MANIPULATION.
https://pandas.pydata.org/docs/

[8]. NUMPY DOCUMENTATION – NUMERICAL COMPUTING WITH PYTHON.
https://numpy.org/doc/

[9]. SCIKIT-LEARN DOCUMENTATION – MACHINE LEARNING IN PYTHON.
https://scikit-learn.org/stable/

[10]. MATPLOTLIB DOCUMENTATION – DATA VISUALIZATION LIBRARY.
https://matplotlib.org/stable/

[11]. OPENPYXL DOCUMENTATION – EXCEL FILE PROCESSING IN PYTHON.
https://openpyxl.readthedocs.io/

[12]. REPORTLAB DOCUMENTATION – PDF REPORT GENERATION.
https://docs.reportlab.com/

[13]. FLASK-LOGIN DOCUMENTATION – USER AUTHENTICATION AND SESSION MANAGEMENT.
https://flask-login.readthedocs.io/

[14]. FLASK-WTF DOCUMENTATION – FORM HANDLING AND VALIDATION.
https://flask-wtf.readthedocs.io/

[15]. WERKZEUG DOCUMENTATION – PYTHON WEB APPLICATION UTILITIES.
https://werkzeug.palletsprojects.com/

[16]. WTFORMS DOCUMENTATION – FORM VALIDATION FRAMEWORK.
https://wtforms.readthedocs.io/

[17]. MYSQL DOCUMENTATION – RELATIONAL DATABASE MANAGEMENT SYSTEM.
https://dev.mysql.com/doc/

[18]. OWASP TOP 10 – WEB APPLICATION SECURITY RISKS.
https://owasp.org/www-project-top-ten/

[19]. HYNDMAN, R. J., & ATHANASOPOULOS, G. – FORECASTING: PRINCIPLES AND PRACTICE (3RD ED.).
OTexts. https://otexts.com/fpp3/

[20]. MCKINNEY, W. – PYTHON FOR DATA ANALYSIS (3RD ED.).
O'Reilly Media, 2022.
https://www.oreilly.com/library/view/python-for-data/9781098104023/

[21]. GÉRON, A. – HANDS-ON MACHINE LEARNING WITH SCIKIT-LEARN, KERAS, AND TENSORFLOW (3RD ED.).
O'Reilly Media, 2022.
https://www.oreilly.com/library/view/hands-on-machine-learning/9781098125967/

APPENDICES

D. DATA FLOW DIAGRAM

The Data Flow Diagram (DFD) represents the movement of information within the Stock Level Prediction and Control Software. It shows how data enters the system, how it is processed by different modules, how information is stored in the database, and how the required results are provided to users. The main activities represented in the system include user authentication, product management, inventory control, supplier and purchase management, warehouse and stock transfer, sales and billing, returns, AI demand prediction, and reporting.

In the system, users provide information through different input forms and module interfaces. Product details are processed through the Product Management module, while stock information is handled through the Inventory Control module. Supplier and purchase information is processed through Supplier and Purchase Management, and warehouse-related information is handled through Warehouse and Stock Transfer functions. Sales, billing, and return information are also entered into the system and processed before the corresponding inventory records are updated.

The processed information is stored in the central database and is used by the analytics and prediction components to generate useful outputs. Historical sales and inventory data are provided to the AI Demand Prediction module to estimate future requirements, while the Automated Reorder module uses the results to generate replenishment recommendations. The processed data is then presented through the dashboard, reports, charts, inventory health indicators, sales analysis, and other outputs, allowing users to monitor the system and make informed inventory decisions.

The stored information is further processed by the Inventory Control, AI Demand Prediction, Automated Reorder, Sales, Warehouse, and Analytics modules. Inventory data is analyzed to identify stock conditions, while historical sales and demand information is used for future demand prediction and reorder recommendations

. The processed information is then presented to users through the dashboard, reports, charts, inventory health status, sales trends, and other outputs. Thus, the DFD shows how data is entered, processed, stored, retrieved, and transformed into useful information for effective inventory monitoring and decision making. The entered data is validated and processed before being stored in the central database. The database acts as the main storage area and allows the different modules to retrieve and update information whenever required.

Supplier and purchase information is processed through Supplier and Purchase Management, and warehouse-related information is handled through Warehouse and Stock Transfer functions. Sales, billing, and return information are also entered into the system and processed before the corresponding inventory records are updated. . The processed data is then presented through the dashboard, reports, charts, inventory health indicators, sales analysis, and other outputs, allowing users to monitor the system and make informed inventory decisions.

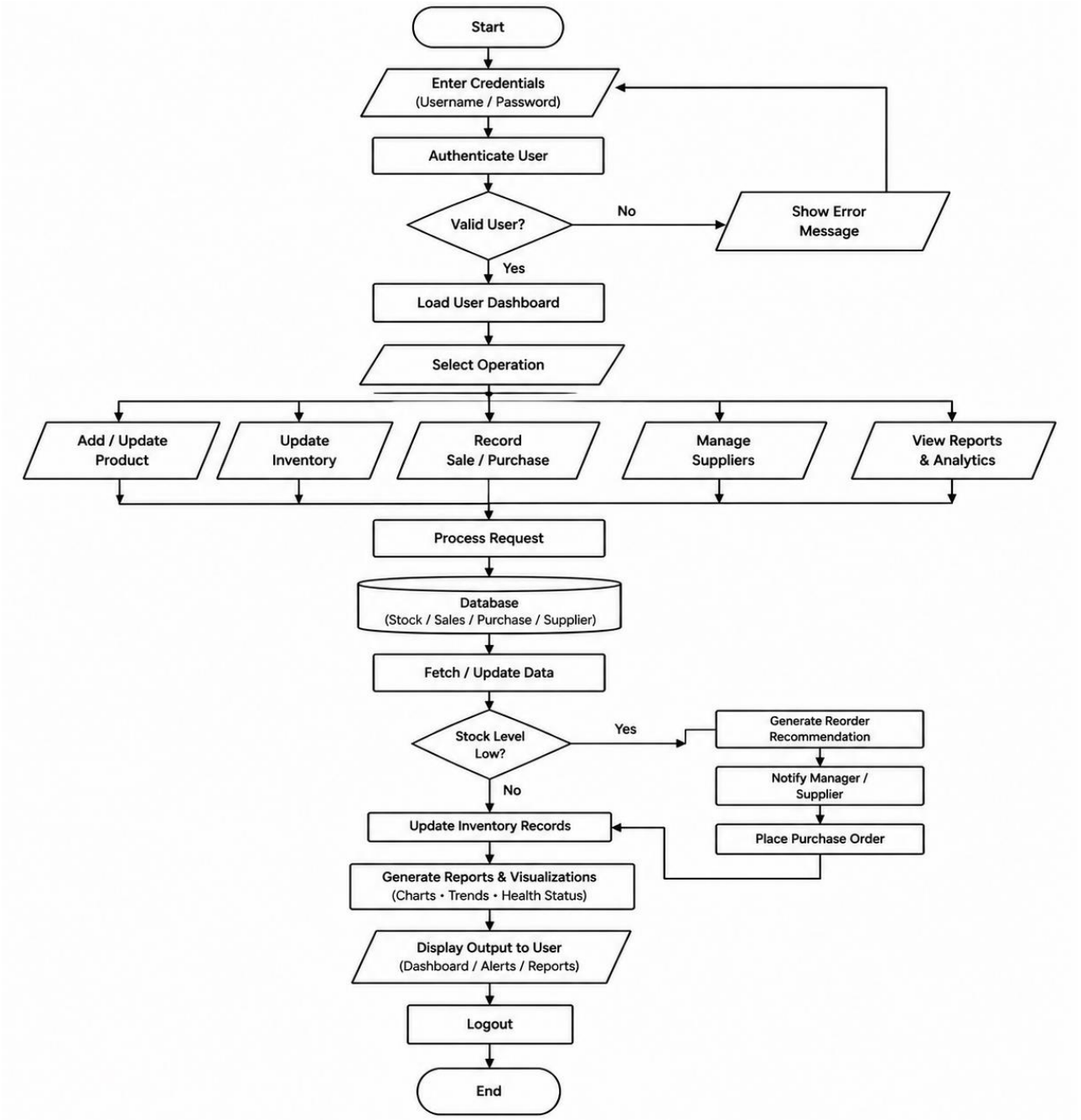


Figure A.2. Data Flow Diagram

E. TABLE STRUCTURE

(i) User Table

Field Name	Data Type	Description
user_id	INT	Unique user identification
name	VARCHAR(100)	User name
email	VARCHAR(100)	User email address
address	VARCHAR(255)	User password
role_id	INT	Assigned user role
created_at	DATETIME	Account creation date

Table 1. User Table

(ii) Role Table

Field Name	Data Type	Description
role_id	INT	Unique role identification
role_name	VARCHAR(50)	Name of the role
description	TEXT	Role description

Table 2. Role Table

(iii) Product Table

Field Name	Data Type	Description
product_id	INT	Unique product identification
product_name	VARCHAR(100)	Product name
supplier_id	INT	Associated supplier
warehouse_id	INT	Associated warehouse
reorder_level	INT	Minimum reorder quantity level

Table 3. Product Table

(iv) Inventory Table

Field Name	Data Type	Description
inventory_id	INT	Unique inventory record
product_id	INT	Associated product
warehouse_id	INT	Associated warehouse
current_stock	INT	Current stock quantity
reserved_stock	INT	Reserved quantity
available_stock	INT	Availabe stock
last_updated	DATETIME	Last inventory update

Table 4. Inventory Table

(v) Supplier Table

Field Name	Data Type	Description
supplier_id	INT	Unique supplier identification
supplier_name	VARCHAR(100)	Supplier name
contact_person	VARCHAR(100)	Contact person
Email	VARCHAR(100)	Supplier email
Phone	VARCHAR(20)	Contact number
location	VARCHAR(100)	Supplier location
Status	VARCHAR(20)	Supplier status

Table 5. Supplier Table

(vi) Warehouse Table

Field Name	Data Type	Description
warehouse_id	INT	Unique warehouse identification
warehouse_name	VARCHAR(100)	Warehouse name
Location	VARCHAR(100)	Warehouse location
total_units	INT	Total stock units
inventory_value	DECIMAL(12,2)	Total inventory value
capacity_utilization	DECIMAL(5,2)	Warehouse capacity usage
status	VARCHAR(20)	Warehouse status

Table 6. Warehouse Table

(vii) Purchase Order Table

Field Name	Data Type	Description
po_id	INT	Unique purchase order ID
supplier_id	INT	Associated supplier
warehouse_id	INT	Receiving warehouse
order_date	DATE	Purchase order date
expected_date	DATE	Expected delivery date
total_amount	DECIMAL(12,2)	Total purchase value
payment_status	VARCHAR(20)	Payment status
delivery_status	VARCHAR(20)	Delivery status
priority	VARCHAR(20)	Order priority

Table 7. Product Order Table

(viii) Sales Table

Field Name	Data Type	Description
sale_id	INT	Unique sales ID
customer_id	INT	Associated customer
user_id	INT	User who processed sale
sale_date	DATETIME	Date and time of sale
total_amount	DECIMAL(12,2)	Total sales value
payment_method	VARCHAR(50)	Payment method
status	VARCHAR(20)	Transaction status

Table 8. Sales Table

(ix) Customer Table

Field Name	Data Type	Description
customer_id	INT	Unique customer identification
Name	VARCHAR(100)	Customer name
Email	VARCHAR(100)	Customer email
Phone	VARCHAR(20)	Customer contact
Address	TEXT	Customer address
status	VARCHAR(20)	Customer status

Table 9. Customer Table

(x) Stock Transaction Table

Field Name	Data Type	Description
transaction_id	INT	Unique transaction ID
product_id	INT	Associated product
warehouse_id	INT	Associated warehouse
transaction_type	VARCHAR(30)	Stock in/out/transfer/return
Quantity	INT	Transaction quantity
transaction_date	DATETIME	Transaction date
created_by	INT	User who created transaction

Table 10. Stock Transaction Table

(xi) Return Table

Field Name	Data Type	Description
return_id	INT	Unique return ID
sale_id	INT	Related sale
product_id	INT	Returned product
customer_id	INT	Customer making return
Quantity	INT	Returned quantity
Reason	INT	Return reason
refund_amount	DECIMAL(10,2)	Refund value
Status	VARCHAR(20)	Return status
return_date	DATETIME	Return date

Table 11. Return Table

(xii) Demand Prediction Table

Field Name	Data Type	Description
prediction_id	INT	Unique prediction ID
product_id	INT	Associated product
predicted_demand	INT	Predicted quantity
forecast_date	DATE	Forecast date
confidence_score	DECIMAL(5,2)	Prediction confidence
model_version	VARCHAR(50)	Prediction model version
created_at	DATETIME	Prediction creation time

Table 12. Demand Prediction Table

(xiii) Reorder Recommendation Table

Field Name	Data Type	Description
recommendation_id	INT	Unique recommendation ID
product_id	INT	Product requiring reorder
suggested_quantity	INT	Recommended order quantity
reorder_date	DATE	Recommended reorder date
Reason	TEXT	Reason for recommendation
Priority	VARCHAR(20)	Critical/High/Medium/Low
Status	VARCHAR(20)	Recommendation status
created_at	DATETIME	Creation date

Table 13. Reorder Recommendation Table

F. SAMPLE CODING

```
import unittest
from app import app
from models import db, User, Product, Supplier, Warehouse, Inventory, Customer, Prediction
class StockSenseTestCase(unittest.TestCase):
    def setUp(self):
        app.config['TESTING'] = True
        self.app = app.test_client()
        with app.app_context():
            db.create_all()
            self.user = User(
                username="testuser",
                email="test@stocksense.com"
            )
            self.user.set_password("test123")
            db.session.add(self.user)
            self.supplier = Supplier(
                name="ABC Suppliers",
                email="abc@gmail.com",
                phone="9876543210"
            )
```

```
db.session.add(self.supplier)
db.session.commit()
self.warehouse = Warehouse(
    name="Main Warehouse",
    location="Coimbatore",
    capacity=1000
)
db.session.add(self.warehouse)
db.session.commit()
self.product = Product(
    id="P001",
    name="Laptop",
    supplier_id=self.supplier.id,
    purchase_price=40000,
    selling_price=50000
)
db.session.add(self.product)
db.session.commit()
self.inventory = Inventory(
    product_id=self.product.id,
    warehouse_id=self.warehouse.id,
```

```
            current_stock=50,
            min_stock=10,
            max_stock=100
        )
        db.session.add(self.inventory)
        db.session.commit()
    def tearDown(self):
        with app.app_context():
            db.session.remove()
            db.drop_all()
    # Test Product
    def test_product(self):
        with app.app_context():
            product = Product.query.get("P001")
            self.assertIsNotNone(product)
            self.assertEqual(product.name, "Laptop")
            self.assertEqual(product.selling_price, 50000)
    # Test Inventory
    def test_inventory(self):
        with app.app_context():
            inventory = Inventory.query.filter_by(
```

```
            product_id="P001"
        ).first()
        self.assertIsNotNone(inventory)
        self.assertEqual(inventory.current_stock, 50)
    def test_low_stock(self):
        with app.app_context():
            inventory = Inventory.query.filter_by(
                product_id="P001"
            ).first()
            inventory.current_stock = 5
            db.session.commit()
            self.assertLess(
                inventory.current_stock,
                inventory.min_stock
            )
    # Test Out of Stock
    def test_out_of_stock(self):
        with app.app_context():
            inventory = Inventory.query.filter_by(
                product_id="P001"
            ).first()
```

```
        inventory.current_stock = 0
        db.session.commit()
        self.assertEqual(
            inventory.current_stock,
            0
        )
    # Test Supplier
    def test_supplier(self):
        with app.app_context():
            supplier = Supplier.query.filter_by(
                name="ABC Suppliers"
            ).first()
            self.assertIsNotNone(supplier)
            self.assertEqual(
                supplier.phone,
                "9876543210"
            )
    # Test Warehouse
    def test_warehouse(self):
        with app.app_context():
            warehouse = Warehouse.query.filter_by(
```

```
            name="Main Warehouse"
        ).first()
        self.assertIsNotNone(warehouse)
        self.assertEqual(
            warehouse.location,
            "Coimbatore"
        )
# Test Customer
def test_customer(self):
    with app.app_context():
        customer = Customer(
            name="Test Customer",
            email="customer@gmail.com",
            phone="9000000000"
        )
        db.session.add(customer)
        db.session.commit()
        saved_customer = Customer.query.filter_by(
            email="customer@gmail.com"
        ).first()
        self.assertIsNotNone(
```

```
            saved_customer
        )
    # Test AI Prediction
    def test_prediction(self):
        with app.app_context():
            prediction = Prediction(
                product_id="P001",
                predicted_sales=75,
                predicted_stock_need=30,
                recommendation="Reorder Stock",
                model_used="Random Forest",
                accuracy_score=0.85
            )
            db.session.add(prediction)
            db.session.commit()
            result = Prediction.query.filter_by(
                product_id="P001"
            ).first()
            self.assertIsNotNone(result)
            self.assertEqual(
                result.model_used,
```

```
            "Random Forest"
        )
        self.assertEqual(
            result.predicted_sales,
            75
        )
# Test User Password
def test_user_password(self):
    with app.app_context():
        user = User.query.filter_by(
            username="testuser"
        ).first()
        self.assertTrue(
            user.check_password("test123")
        )
        self.assertFalse(
            user.check_password("wrong123")
        )
# Test Product API
def test_product_api(self):
    response = self.app.get(
```

```
        '/api/products'
    )
    self.assertEqual(
        response.status_code,
        200
    )
# Test Dashboard API
def test_dashboard_api(self):
    response = self.app.get(
        '/api/general/dashboard'
    )
    self.assertEqual(
        response.status_code,
        200
    )
# Test Inventory API
def test_inventory_api(self):
    response = self.app.get(
        '/api/inventory'
# Test Supplier API
def test_supplier_api(self):
```

```
        response = self.app.get(
            '/api/suppliers'
        )
        self.assertEqual(
            response.status_code,
            200
        )
    # Test Prediction API
    def test_prediction_api(self):
        response = self.app.get(
            '/api/predictions'
        )
        self.assertEqual(
            response.status_code,
            200
        )
if __name__ == '__main__':
    unittest.main()
```

G. SAMPLE INPUT

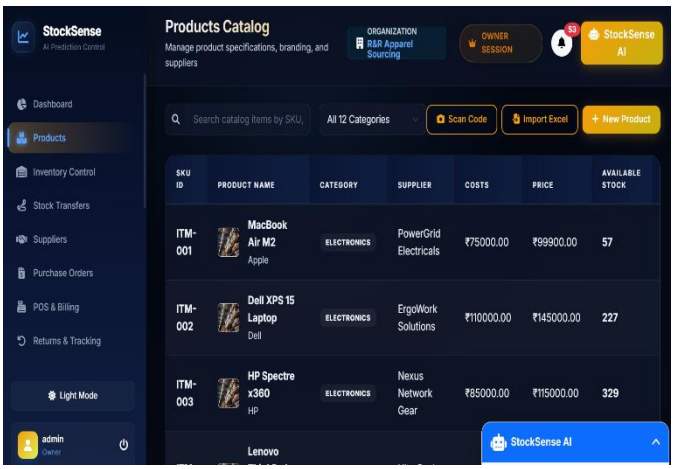


Fig 18. Product Entry

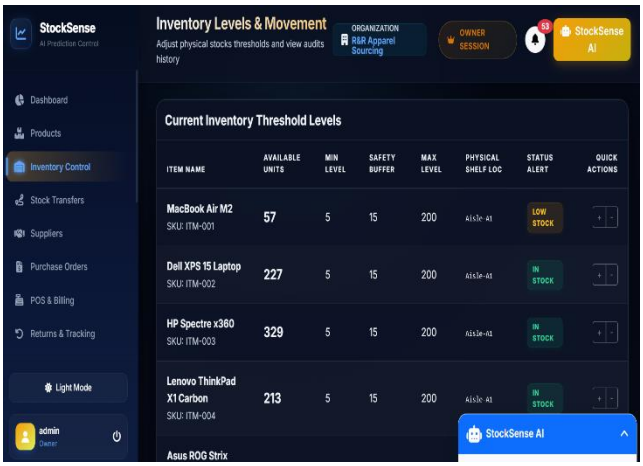


Fig 19. Inventory Control

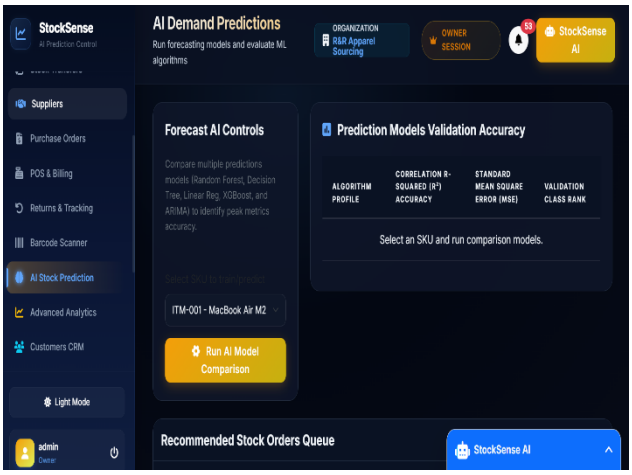


Fig 20. AI Stock Prediction

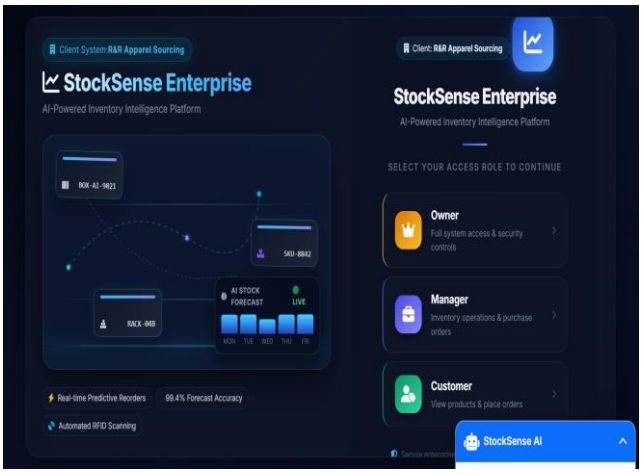


Fig 21. Login Screen



Fig 22. Owner Dashboard

H. SAMPLE OUTPUT

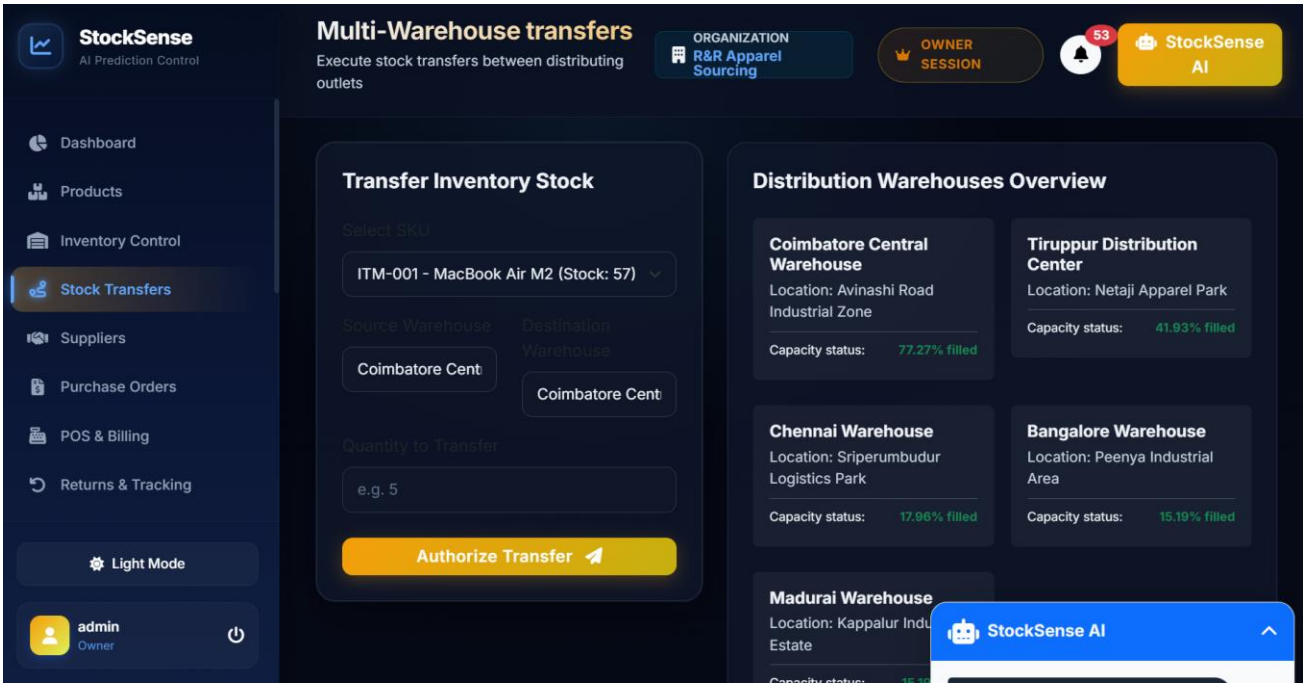


Fig 23. Stock Transfer

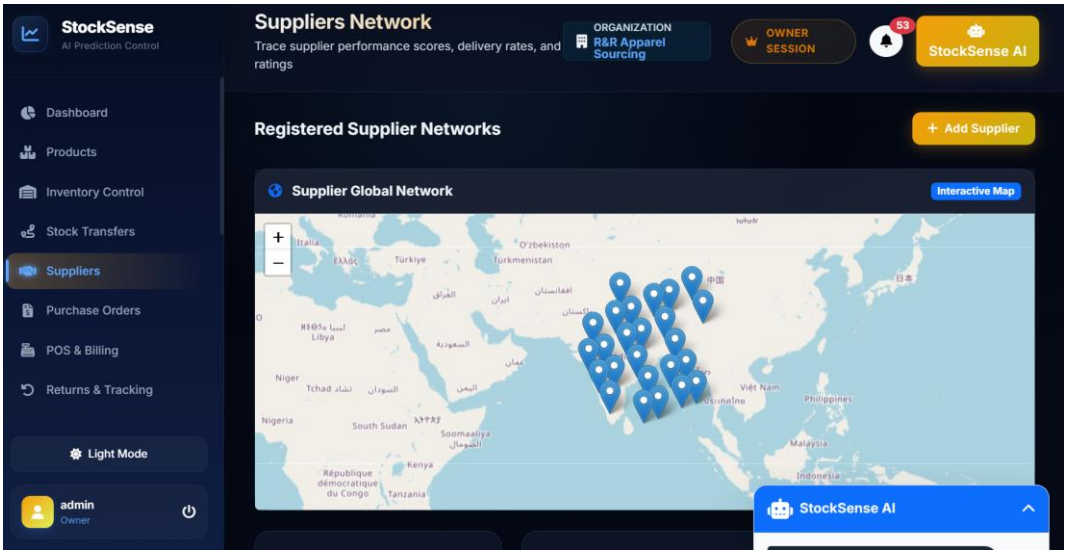


Fig 24. Suppliers Screen

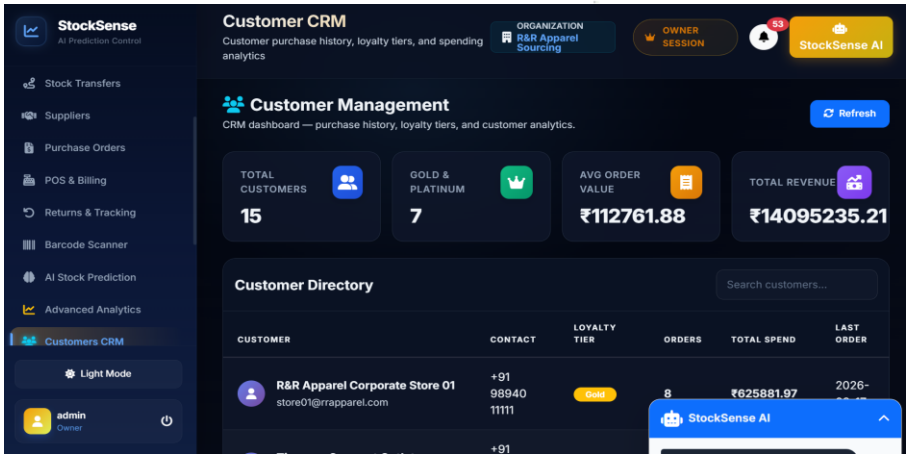


Fig 25. Customer CRM

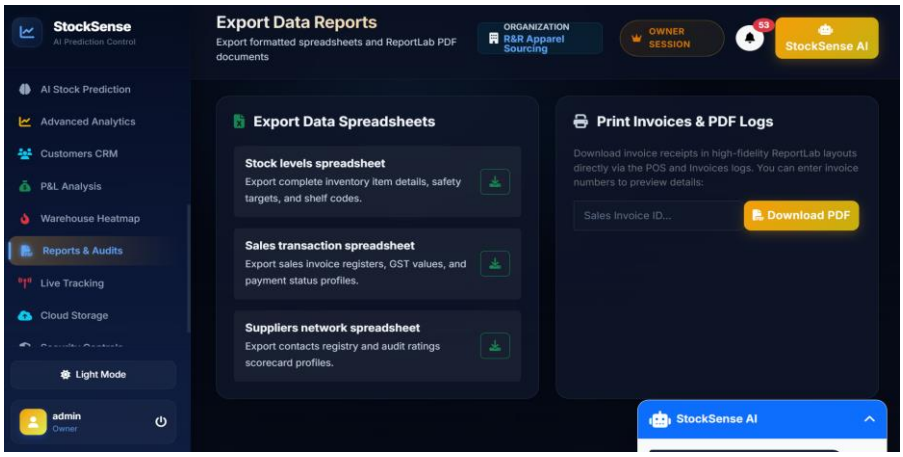
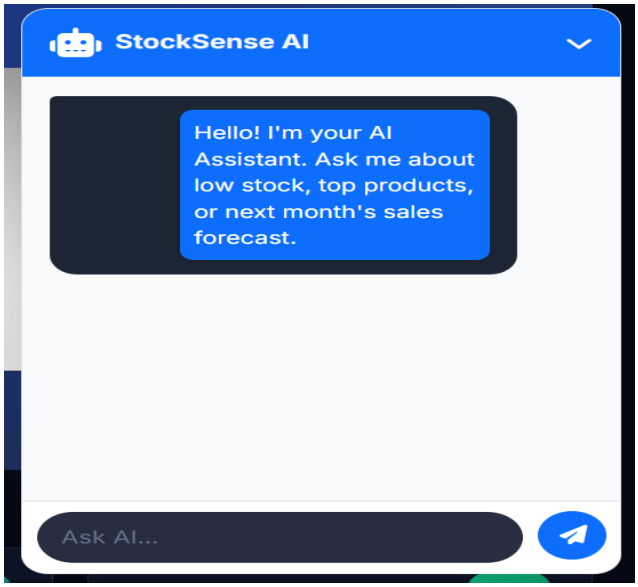


Fig 26. Reports & Audits



R

Fig 27. Chatbot Screen



Fig 26. P&L Analysis